LongiTumorSense: Longitudinal AI for Brain Tumor Detection and Progression Analysis

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ABSTRACT

Among many other diseases one of the deadliest disease type is Brain Tumors. Glioma is one of the Primary Brain Tumors which is often fatal. According to the World Health Organization (WHO), gliomas account for approximately 30% of all brain tumors and 80% of Malignant Brain Tumors, with an estimated 100,000 new cases diagnosed globally each year. Due to the aggressive growth and complicated nature of tumors, it is difficult to analyze MRI data and forecast tumor progression.

As brain tumor diagnosis and monitoring continue to face significant difficulties in the age of improved medical imaging and machine learning. In order to overcome these obstacles, the suggested system creates an online platform that enables users to upload MRI scans in order to identify brain tumors (Glioma), monitor their chance of recurrence, forecast their chance of survival and get treatment recommendations.

The system uses free GPU resources like Goggle Colab to train distinct models for tumor recognition, segmentation and clinical outcome prediction using the publicly available MU-Glioma-Post dataset. To Guarantee ethical use, an admin panel limits access to verified users by confirming through credentials, degree or research papers. Two Primary features are an Explore button that generates tumor detection, survival and treatment reports for single MRI upload and a Compare button that includes progression or recurrence analysis for numerous MRI uploads. This system uses open-source tools and frameworks to assure cost and scalability as an effort to democratize access to advanced tumor analysis especially in areas with limited resources like Pakistan.

CERTIFICATE

Dated: _____

Final Approval

It is certified that project report titled "**LongiTumorSense"** submitted by **Abdullah, Ehtasham-ul-Haq and Hassan Ali** for the partial fulfillment of the requirement of **"Bachelor's Degree in Software Engineering"** is approved.

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DECLARATION

We hereby declare that our dissertation is entirely our work and genuine/original. We understand that in case of discovery of any PLAGARISM at any stage, our group will be assigned an F(FAIL) grade and it may result in withdrawal of our Bachelor's degree

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TABLE OF CONTENTS

CHAPTER **PAGE**

CHAPTER 1 INTRODUCTION 1

1.1 Introduction 2
1.2 Problem Description 2
1.3 Project Objective 3
1.4 Project Scope 3
1.5 Developed System 3
1.5.1 Features 4
1.5.2 Expected Result 4
1.5.3 Possible Extensions 4
1.5.4 Limitations 4
1.6 Modules 4
1.6.1 Admin Panel module 4
1.6.1.1 Flow chart of Admin Panel 5
1.6.2 Tumor Detection Module 6
1.6.2.1 Flow chart of Tumor Detection 6
1.6.3 Progression/Recurrence Analysis Module 7
1.6.3.1 Flow chart of progression/Recurrence Analysis 7
1.6.4 Clinical Outcome Prediction Module 8
1.6.4.1 Flow Chart of Clinical Outcome Prediction 8
1.7 Process Flow 9
1.8 Feasibility Study 10
1.8.1 Technical Feasibility 10
1.8.2 Economic Feasibility 10
1.8.3 Legal and Regulatory Feasibility 10

1.9 Risk Involved 11
1.9.1 Data Security and Privacy Concerns 11
1.9.2 User Verification 11
1.9.3 System Performance 11
1.10 Solution Application Areas 11
1.11 Methodology 12
1.12 Tools and Technologies 13
1.12.1 Software 13
1.12.2 Hardware 13
1.13 Team Members Expertise 14
1.14 Milestones 15
1.15 Summary 16

CHAPTER 2 BACKGROUND AND EXISTING SYSTEMS 17

2.1 Introduction 18
2.2 Existing Studies 18
2.2.1 BraTS-Dataset Based System 18
2.2.1.1 Features 18
2.2.1.2 Limitations 18
2.2.2 CBICA's Image processing portal 18
2.2.2.1 Features 18
2.2.2.2 Limitations 19
2.2.3 DeepMedic 19
2.2.3.1 Features 19
2.2.3.2 Limitations 19
2.2.4 MedSeg 19
2.2.4.1 Features 19
2.2.4.2 Limitations 19
2.2.5 TumorNet 20

2.2.5.1 Features....20
2.2.5.2 Limitations....20
2.3 Comparison between proposed system and existing systems....20
2.4 Summary....20

CHAPTER 3 REQUIREMENT SPECIFICATION....21

3.1 Introduction....22
3.2 Interface Requirements....22
3.2.1 User Interface....22
3.2.2 Hardware Interfaces....22
3.2.3 Software Interfaces....22
3.3 Functional Requirements....23
3.3.1 Registration of New User....23
3.3.2 Forget password....24
3.3.3 Create Multiple Patients Profiles....24
3.3.4 Add Clinical Data for Patient....24
3.3.5 Upload MRI scans for detection....25
3.3.6 Generate Progression Report....25
3.3.7 Compare MRI's....25
3.3.8 Save generated report....26
3.3.9 Delete Patient records....26
3.4 Non-Functional Requirements....27
3.4.1 Performance....27
3.4.2 Security....27
3.4.3 Availability....27
3.4.4 Usability....27
3.4.5 Reusability....28

3.4.6 Flexibility 28
3.4.7 Maintainability 28
3.4.8 Portability 28
3.5 Use Case Modeling 29
3.6 Use Case Description 30
3.6.1 Use case: User Registration 30
3.6.2 Use case: Log In 31
3.6.3 Use case: Reset password 32
3.6.4 Use case: Manage Patients Profile 33
3.6.5 Use case: View Patient Medical Records 34
3.6.6 Use case: Upload MRI scans 35
3.7 Resource Requirement 36
3.7.1 Hardware Requirements 36
3.7.2 Tools/Technology 36
3.8 Human Effort 37
3.9 Database Requirements 37
3.10 Summary 37

CHAPTER 4 SYSTEM MODELING 38

4.1 Introduction 39
4.2 (4+1) View Model 39
4.3 Logical View 40
4.3.1 Class Diagram 40
4.4 Process View 40
4.4.1 System Activity Diagram 41
4.4.2 Sequence Diagram 42
4.4.2.1 Registration Sequence Diagram 42
4.4.2.2 Login Sequence Diagram 43

4.4.2.3 Explore Sequence Diagram 44
4.4.2.4 Compare Sequence Diagram 45
4.4.3 ER Diagram 46
4.5 Development View 47
4.5.1 Component Diagram 47
4.6 Physical View 48
4.6.1 Deployment Diagram 48
4.7 System Prototype 49
4.8 Summary 53

REFERENCES 54

LIST OF FIGURES

Figure **Caption** **Page**

1.1 Flowchart of Admin Panel 5

1.2 Flowchart of Tumor Detection 6

1.3 Flowchart of Progression Analysis 7

1.4 Flowchart of clinical outcome prediction 8

1.5 System process flow 9

1.6 Incremental Model 12

1.7 Gant Chart to illustrate project completion 15

3.1 System Use Case Diagram 29

4.1 (4+1) View Model 39

4.2 System Class Diagram 40

4.3 System Activity Diagram 41

4.4 Registration Sequence Diagram 42

4.5 Login Sequence Diagram 43

4.6 Explore Sequence Diagram 44

4.7 Compare Sequence Diagram 45

4.8 Entity Relationship Diagram for LongiTumorSense 46

4.9 System Component Diagram 47

4.10 Deployment Diagram 48

4.11 System Landing page 49

4.12 Login screen 50

4.13 Registration screen 50

4.14 Doctor Interface 51

4.15 Admin Interface 51

4.16 Adding Patient Clinical Data 51

4.17 Generated Report 52
4.18 Treatment Recommendation 52

LIST OF TABLES

Table **Caption** **Page**

1.1 Tools and Technologies....13

1.2 Team Members Expertise....14

2.1Comparative Analysis between Proposed system and Existing systems....20

3.1 User Registeration....23

3.2 Forget password....24

3.3 Patient profiles creation....24

3.4 Adding Clinical data of Patient....24

3.5 Uploading MRI scans....25

3.6 Generate Progression Report....25

3.7 MRI's Comparison....25

3.8 Save Generated Report....26

3.9 Logout System....26

Use case description for User Registration....30

3.11 Use case description for User Login....31

3.12 Use case description for Password reset....32

3.13 Use case description for Patient creation....33

3.14 Use case description for Medical Records History....34

3.15 Use case description for MRI Upload....35

3.16 System Requirements....36

3.17 Human Effort....37

CHAPTER 1
INTRODUCTION

1.1 Introduction

Brain tumor detection and treatment have changed dramatically because of quick development of Artificial Intelligence and medical imaging. Although magnetic resonance imaging (MRI) is essential for identifying and tracking brain malignancies, radiologist's manual interpretation of MRI data is laborious and subject to error. Delay in tumor detection and treatment planning are made worse in places like Pakistan by lack of access to advanced diagnostic equipment and knowledge. Current systems frequently concentrate on particular areas, such as tumor identification and segmentation, but they do not provide a thorough study of progression, recurrence, survival rates and recommended treatments.

LongiTumorSense a web-based system that works with trained machine learning models for tumor (Glioma) detection, segmentation, progression/recurrence analysis and clinical outcome prediction using the MU-Glioma-Post Dataset. By verifying educational documents, certifications or research documents, the system's protected admin panel restricts access exclusively to verified medical professionals such as physicians, radiologists and researchers thereby guarantying ethical use. This system is intentionally designed only for authenticated doctors and medical experts, meaning that ordinary users or patients cannot access or operate it. To improve use there are two buttons : first Explore for single MRI uploads to find cancers, forecast survival and recommend therapies and second button Compare for numerous MRI uploads to evaluate progression and recurrence . The system guarantees cost and accessibility for certified and authorized experts by utilizing open-resource frameworks and free resources like Google Colab and Kagle.

1.2 Problem Description

Brain tumor diagnosis and treatment planning remain critical challenges in modern healthcare era, specifically where resources are constrained such as Pakistan. MRI provides a gold standard for detection of brain tumors but manual interpretation of MRI scans by radiologists is resource consuming, prone to error and often delayed due to limited diagnostic infrastructure. Although some tools exist for brain tumor analysis but they tend to ignore progression tracking, survival prediction, therapy recommendations and ethical access restriction in favor of detection or segmentation. Usually requiring costly computational resources, these systems are in accessible to non-experts or areas with fewer resources. Delays in diagnosis and treatment commonly need a secure yet affordable system

that integrates detection, analysis and predictions using open source datasets and tools within an intuitive web interface.

1.3 Project Objective

The primary goal of the project is a web-based system “LongiTumorSense” designed to make advanced brain tumor analysis affordable and accessible to healthcare professionals working in underserved regions, built by using publicly available the MU-Glioma-Post dataset, the system focuses on identifying Brain tumors (Glioma) from MRI scans, monitor their progression or recurrence, forecast survival rates and recommend treatments. To ensure ethical and responsible use of platform, “LongiTumorSense” includes a secure admin panel that will limit access to only verified medical professionals including radiologists, physicians and qualified medical researchers. Ensuring responsible use and optimizing accessibility for health care providers in areas with limited resources.

1.4 Project Scope

A secure online platform for thorough study of brain tumors, particularly gliomas, is what the created technology “LongiTumorSense” is intended to be. It seeks to streamline and improve the diagnostic process by enabling verified medical professionals to Upload MRI scans and obtain automated insights into the presence and classification of tumors(Gliomas) , providing tracking of tumor progression and recurrence along with survival rate predictions and treatment recommendations. It uses scalable deployment with free resources and restricted access through an admin panel for approved medical experts to guarantee responsible and moral use.

1.5 Developed System

LongiTumorSense is a secure web-based system that allows verified medical practitioners to upload MRI scans for analyzing brain tumors (Glioma).The MU-Glioma-Post dataset is used by the system to train distinct models for clinical outcome prediction, tumor detection, segmentation and progression/recurrence analysis. By confirming degrees, certifications or research papers. An admin panel authenticates users and ensures that only physicians, radiologists or researchers have access to the system, thereby addressing ethical concerns. “Explore” and “Compare” two buttons, serve distinct purposes: One MRI uploads for clinical predictions and detection and Several MRI uploads for progress and thorough analysis. The system is developed using free technologies such as Tensor Flow, React.js

and GPU support from platforms like Kaggle and Goggle Colab to ensure cost effective development and deployment.

1.5.1 Features

- Admin Panel and secure web based application for user authentication.
- User-friendly interface consisting “Explore” and “Compare” buttons.
- Segmentation and identification of tumors.
- Analysis of recurrence and progression with Multiple MRI’s.
- Prediction of survival rates and recommendations for treatments.

1.5.2 Expected Result

- Precise Tumor segmentation and identification
- Trustworthy monitoring of progression and recurrence.
- Exact forecasts for survival and therapy.
- Access that is ethical and safe for professionals that have been authenticated.
- An accessible platform for areas with limited resources.
- Improved analysis of MRI scans by the use of longitudinal dataset.

1.5.3 Possible Extensions

Should we meet our time constraints, we plan to generate more detailed analysis on tumor characterization by incorporating radiomic features offering both patient and doctors a clear data-driven understanding of health outcomes. We also intend to integrate our system with hospitals to ensure scalability and accessibility for both patient and doctors across different regions

1.5.4 Limitations

- For operation and access, it requires a steady internet connection.
- The system can only handle MRI scans, no camera format images can be processed

1.6 Modules

Following are the modules of “LongiTumorSense” along with their key features:

1.6.1 Admin Panel module

This Module Controls user authentication by permitting sign-ups with uploaded documents (research papers, degrees and certifications) and admin verification to guarantee that the system is only accessible by medical professionals.

1.6.1.1 Flow chart of Admin Panel

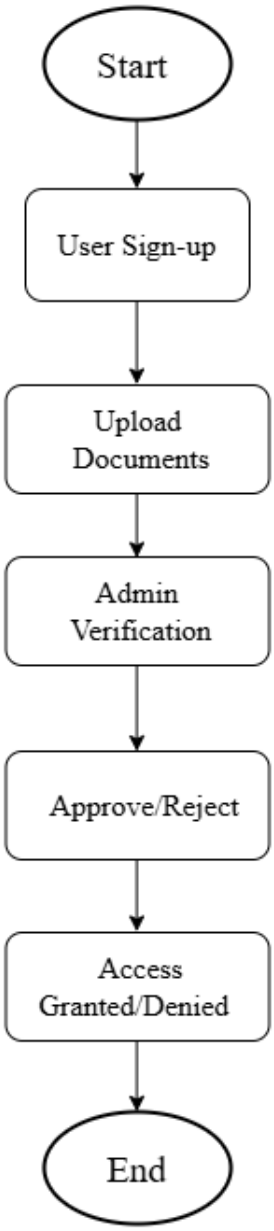


Figure 1.1 Flowchart of Admin Panel

1.6.2 Tumor Detection Module

This module uses a convolutional neural network (CNN) trained on the MU-Glioma-Post dataset to process one or more MRI scans in order to identify and segment brain tumors.

1.6.2.1 Flow chart of Tumor Detection

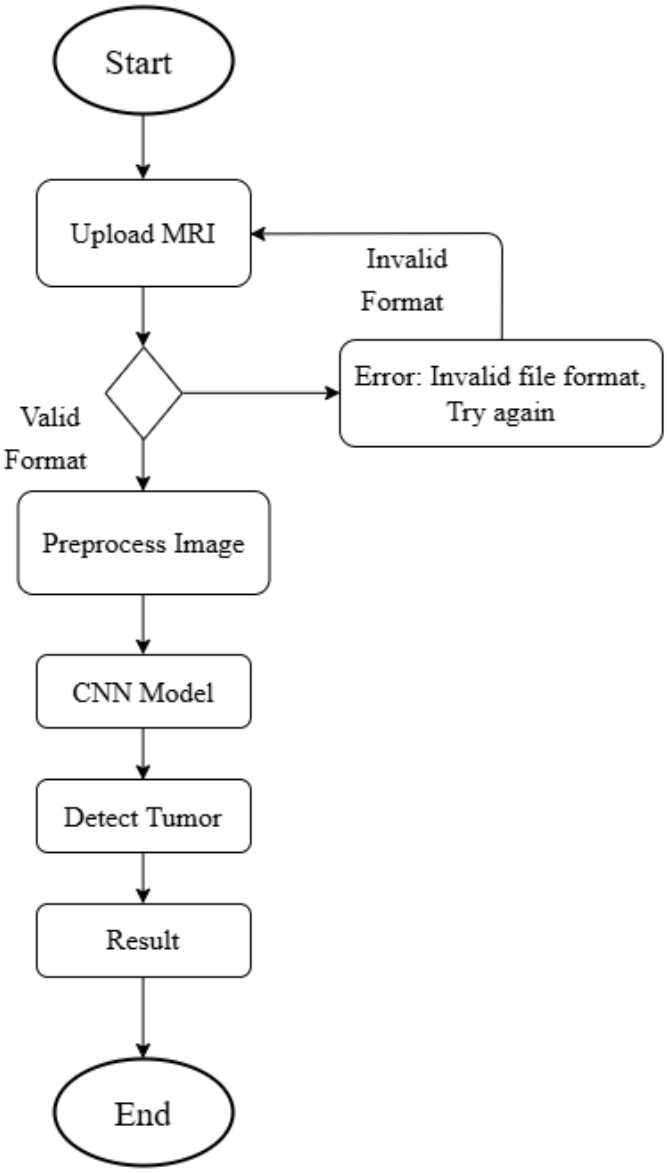


Figure 1.2 Flowchart of Tumor Detection

1.6.3 Progression/Recurrence Analysis Module

Through feature comparison across different time stamps, this module examines several MRI scans to monitor tumor growth or recurrence

1.6.3.1 Flow chart of progression/Recurrence Analysis

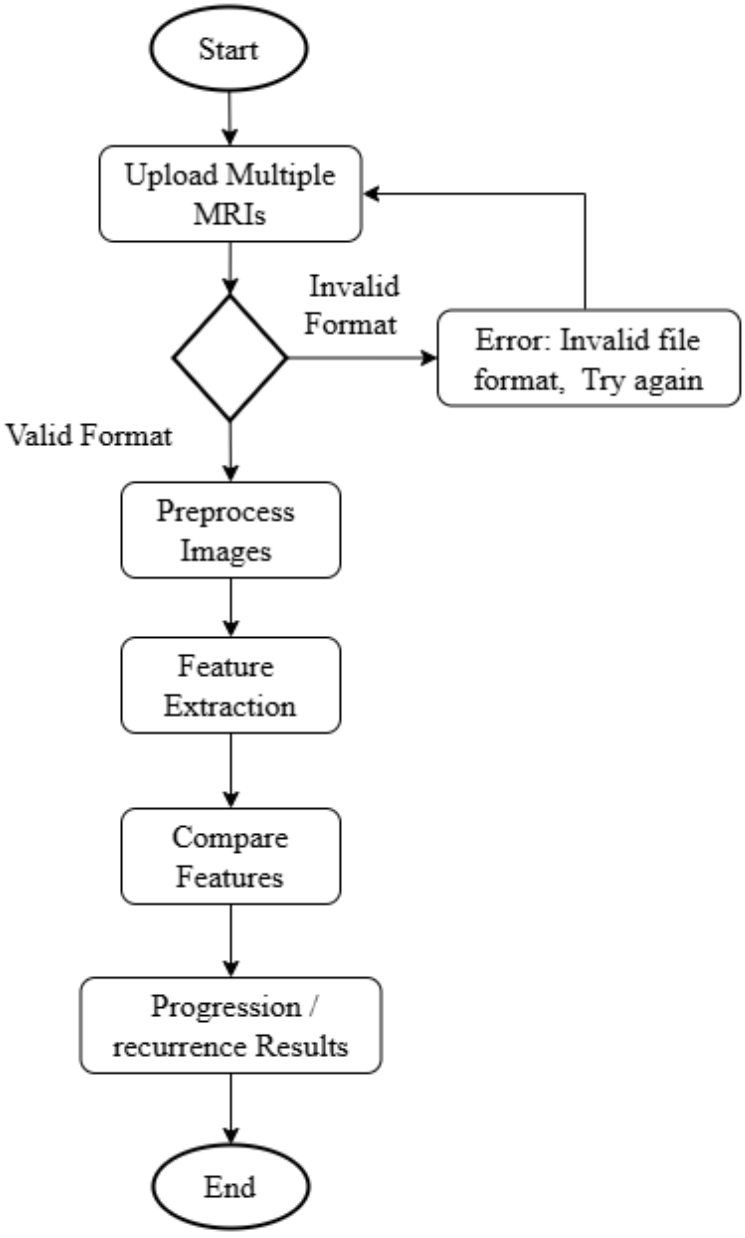


Figure 1.3 Flowchart of Progression Analysis

1.6.4 Clinical Outcome Prediction Module

This module uses clinical information from the MU-Glioma-Post dataset and tumor features to forecast survival rates and recommend therapies

1.6.4.1 Flow Chart of Clinical Outcome Prediction

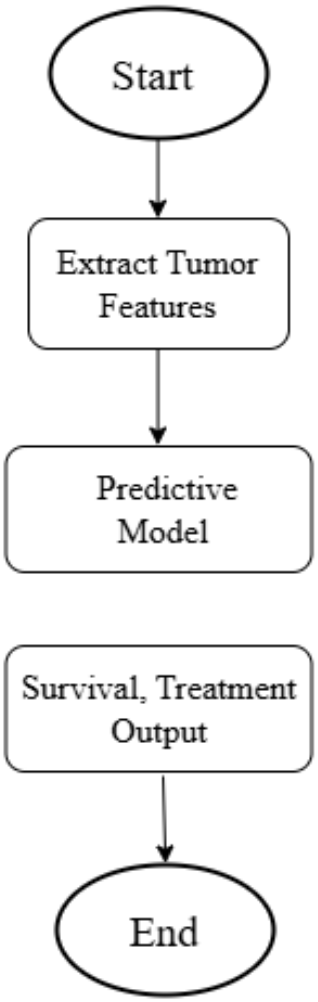


Figure 1.4 Flowchart of clinical outcome prediction

1.7 Process Flow

This process flow shows how the patient interacts with the mobile application. Figure illustrates the sequence of flow from the patient perspective.

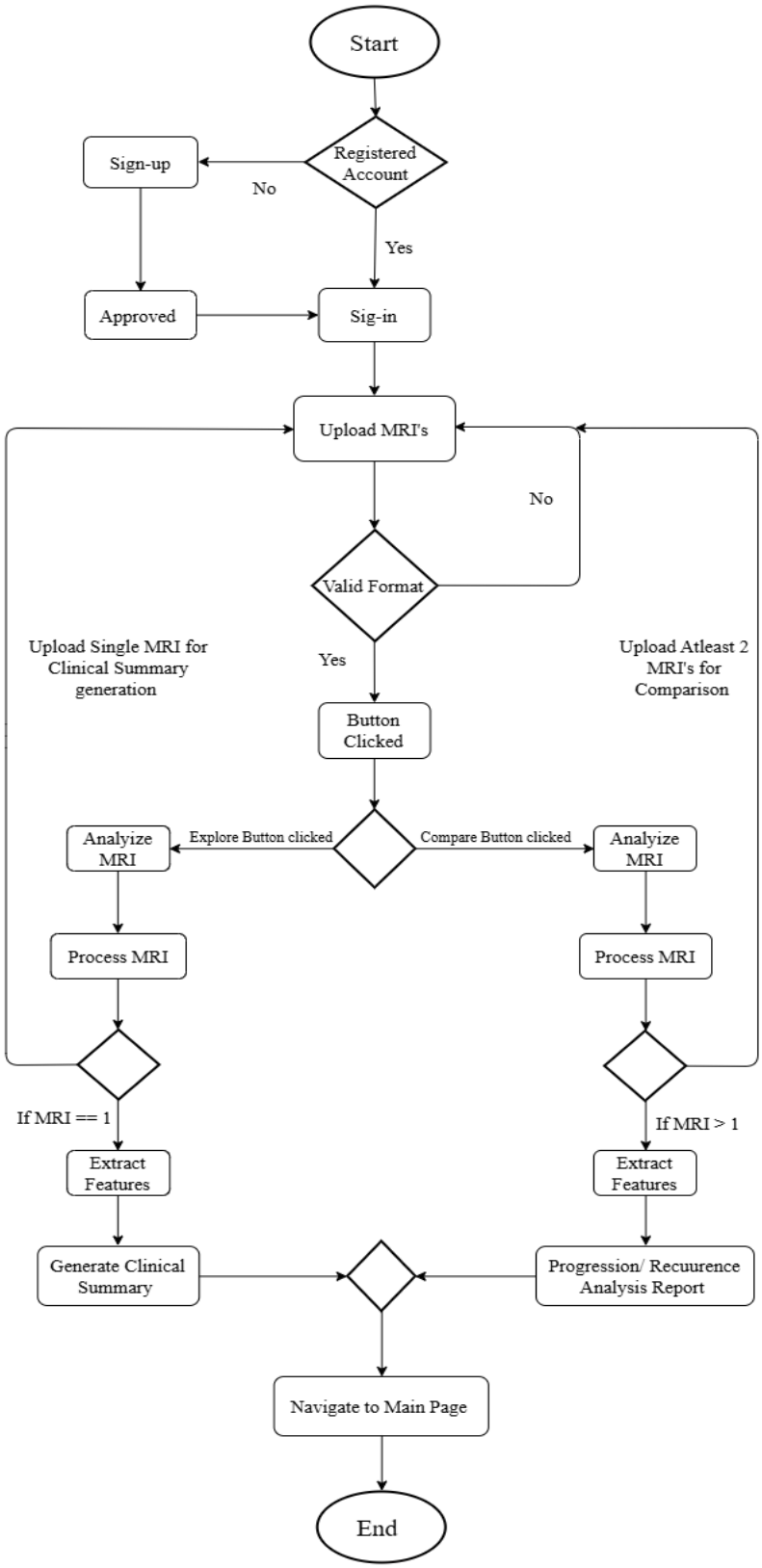


Figure 1.5 System process flow

1.8 Feasibility Study

A Feasibility study assesses the viability and practicality of the Developed system. It is the systematic analysis and evaluation of the success of the Developed project. Following is the feasibility study of the “LongiTumorSense”.

1.8.1 Technical Feasibility

The technical feasibility of “LongiTumorSense” is clear. Our team possesses the fundamental knowledge required for the development of this system like Web Development, Machine learning , Deep learning ,UI/UX designing, database management and Image processing. This ensures that we have the necessary skills to develop the system. The software tools, datasets and frameworks that we have selected for development, model training and designing are freely available to download and use, these tools will provide us with necessary resources required for the development of our system.

1.8.2 Economic Feasibility

The Economic Feasibility of “LongiTumorSense” is financially viable and cost-effective. The economic feasibility of our system focuses on the expenses that are directly related to the development cost of the system. We won’t have to face any major expenses related to hardware as all development will be done on our pc’s. The software and frameworks that we will be using are mostly freely available to download and use, however if some tool require subscription fee we would need to account for that expense. Overall, in terms of software we donot anticipate any significant expense. The primary expense will be the documentation costs , which involve the expense of priniting project-related documents and the hosting cost for system deployment.

1.8.3 Legal and Regulatory Feasibility

The legal and regulatory feasibility of “LongiTumorSense” is carefully navigated to comply with relevant healthcare and data protection laws. The system will rigorously adhere to the Pakistan Electronic Transactions Ordinance 2002, ensuring the legal recognition and protection of electronic data transactions. Compliance with the Data Protection and Privacy Las in Pakistan if enacted will be a priority, emphasizing the secure collection, storage and processing of patients MRI’s and clinical data. To address ethical

considerations, the system will align with Pakistan Medical and Dental Council (PMDC) guidelines, emphasizing principles such as patient autonomy and confidentiality.

1.9 Risk Involved

Following are some key risks that can potentially cause our system to fail:

1.9.1 Data Security and Privacy Concerns

The occurrence of data breaches and privacy violations can lead to unauthorized access to the patient's data resulting in data loss and compromised data integrity, which can cause loss of users trust in the application creating a domino effect leading to failure of the system in protecting user's private data.

1.9.2 User Verification

There is a Possibility of missing critical documents during professional verification unauthorized individuals gaining access if verification is incomplete

1.9.3 System Performance

System depends on stable internet connection for accurate results, could cause delays, slow performance or improper results in poor-bandwidth environments

1.10 Solution Application Areas

LongiTumorSense can be applied and utilized in the following areas:

- **Clinical Diagnosis:**
  Assisting health care professionals in accurate and efficient diagnosis of brain tumors (gliomas) through advanced image processing and access to relevant health care information.
- **Organizations conducting gliomas studies:**
  Can be used by organizations that are conducting studies in brain tumors specifically gliomas or for research purposes.
- **Hospital Radiology Department:**
  Streamline the interpretation of MRI's in the hospitals radiology department, facilitating the rapid diagnosis of brain tumor in clinics and hospitals.

1.11 Methodology

The methodology chosen for the project is “Incremental model” which permits development in stages (detection, progression, clinical prediction and admin panel) to guarantee stability and iterative enhancement.

We have used incremental model for this system for the following reasons:

- Allows accommodating changes with ease, making the system development flexible and capable of evolving.
- Allows parallel deployment and testing of different increments, leading to efficient and fast development.
- Includes early and continuous user feedback, which ensures that the system is according to the user’s requirements, leading to the identification and resolution of issues and enhancing the functionality of the system.
- Provides a well-organized framework for project management, making it easier to plan, monitor and control development.

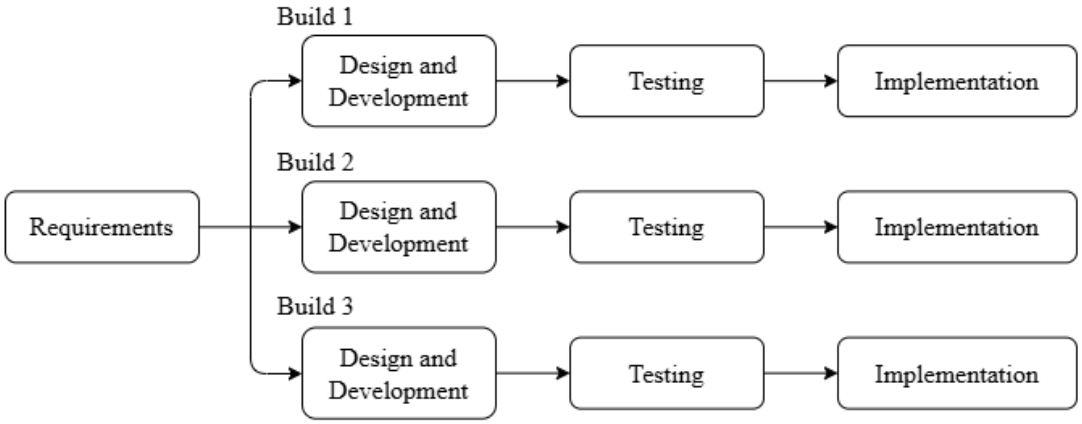


Figure 1.6 Incremental Model

1.12 Tools and Technologies

Following tools and technologies will be required for the development of the system:

1.12.1 Software

Software tools required for the development of LongiTumorsense:

- Python (PyTorch)
- React JS
- Express Js
- BraTS-Style U-Net,3D U-Net, Enhanced 3D U-Net, DenseNet-121 , ResNet-50, EfficientNet-B4
- Jupyter Notebook/Google Colab
- NIfTI(.nii),CSV, PNG, JPEG
- Pandas,NumPy
- Matplotlib, Seaborn,ITK-SNAP
- NIBabel,SimpleITK

1.12.2 Hardware

- PC or Laptop core i5/i7 5th generation or above.
- 500 GB or above storage
- 8 GB ddr3 or above RAM

Table 1.1 Tools and Technologies

Tools and Technology	Explanation
PyTorch	Building machine learning and deep learning models
React JS	Frontend development for user interaction
Express JS	Backend server for integration of response with interface
UNet, 3DUNet, DenseNet-121, ResNet-50	Implementing medical image segmentation and classification models
Google Colab	Collaborative coding and running experiments in cloud environments
NIBabel, SimpleITK	Medical Image processing and file format handling

1.13 Team Members Expertise

Our team is composed of skilled professionals with diverse expertise in software development, UI/UX design, database management and mobile application development ensuring a comprehensive approach to the project . Each member brings proficiency in programming languages such as JavaScript, Python and web technologies like React Tailwind Css and Bootstrap. Their knowledge extends to design architecture, UML modeling and system documentation allowing a structured and efficient project execution. Additionally, expertise in project management ensures that all aspects of the system are well managed. The combination of these technical skills and problem solving abilities enables our team to develop a high quality, user-friendly and scalable solution, addressing key challenges in Tumor Detection effectively. Table 1.2 shows Team Members Expertise.

Table 1.2 Team Members Expertise

Name	Expertise
Abdullah	Web Development (node.js, Express.js MongoDB), UI/UX Design, Python, Documentation, UML Diagrams and Modeling, Project Management.
Ehtasham Ul Haq	Python, Machine Learning, Web development (React js, Tailwind css), Documentation, Team Leadership, Project Management.
Hassan Ali	Python, Deep Learning, Data preprocessing, documentation, Project Management, Web development (html, css, JavaScript).

1.14 Milestones

A Gant chart is a management tool that visually represents tasks over time, providing a clear overview of scheduled activities, dependencies and timelines to ensure the systematic development of LongiTumorSense. This Graphical representation helps track progress, meet deadlines and maintain an efficient workflow throughout the project. Figure 1.7 illustrates the Gantt chart for LongiTumorSense Brain Tumor Detection System.

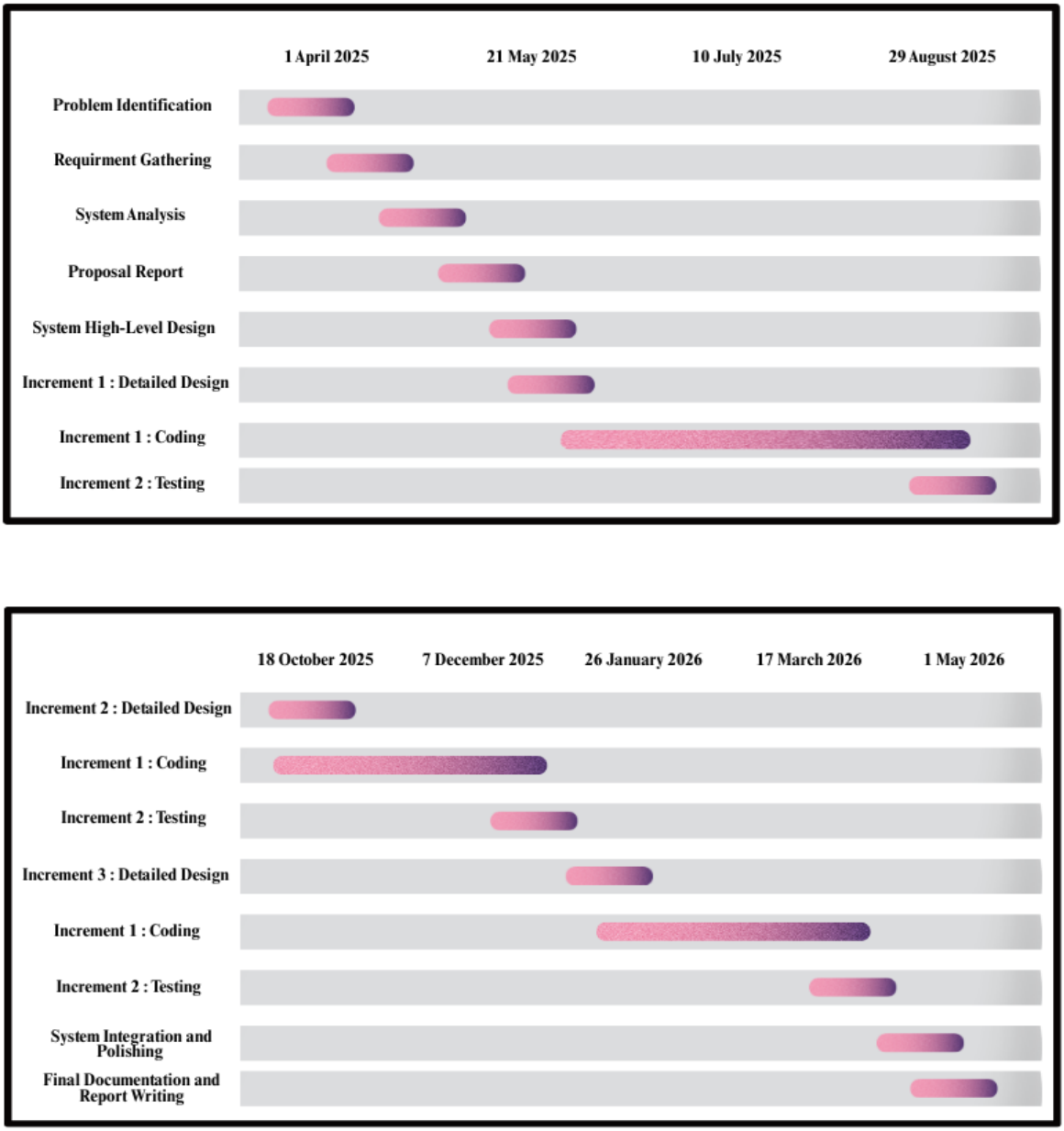


Figure 1.7 Gant Chart to illustrate project completion

1.15 Summary

In this chapter, we have discussed about our Developed system its modules and key features, highlighting the problem, objective of our system, motivation behind developing the system and scope of our project. We have shown the flow of our system along with the process model that we are going to follow to develop this system. This Chapter highlights the possible risks that could cause system failures along with the tools and technology required to develop the system.

CHAPTER 2
BACKGROUND AND EXISTING SYSTEMS

2.1 Introduction

This chapter discusses the existing systems related to our system. As the current scope of existing systems are just segmentation and detection with the use of MRI datasets like BraTS (Brain Tumor Segmentation).But few combine combined treatment recommendations, survival prediction, progression tracking and restricted access for moral use into a single platform. Notable systems are analyzed below, along with their detailed comparison with our Developed system.

2.2 Existing Studies

Following are the existing systems on Brain Tumor detection and their brief explanation:

2.2.1 BraTS-Dataset Based System

Systems based on BraTS (Brain Tumor Segmentation) dataset are frequently used to test tumor segmentation methods. These technologies use multimodal MRI scans to automatically identify tumors.

2.2.1.1 Features

- Segmentation of tumor sub-region with high accuracy (e.g. enhancing tumor, edema)
- Research can be conducted, as it is an open source dataset.
- Supports U-NET and other deep learning models.

2.2.1.2 Limitations

- Focusses only on segmentation.
- Demands a large amount of processing power, frequently expensive GPU's.
- Limited real world variability, dataset lacks diverse clinical imaging conditions, which may reduce model robustness in non-standard environments.

2.2.2 CBICA's Image processing portal

The CBICA portal provides cloud-based tools for MRI scan based tumor segmentation and analysis.

2.2.2.1 Features

- Web-based MRI scan uploading interface
- Automated extraction of radiomic features and segmentation
- Encourages clinical application not research.

2.2.2.2 Limitations

- Limited to feature extraction and segmentation, not survival or progression analysis
- Requires the creation of account but no expert verification is done.
- Large datasets may result in delayed processing.

2.2.3 DeepMedic

DeepMedic is an open source program that uses deep learning to segment 3D brain Tumors.

2.2.3.1 Features

- Effective multi-model MRI scan segmentation
- Designed with reduced memory usage in mind.
- Both open source and adaptable.

2.2.3.2 Limitations

- No tracking of progression or recurrence.
- Lack modules for suggesting treatments or predicting clinical outcomes.
- For ethical reasons, there is no access control.

2.2.4 MedSeg

A web-based program for segmenting medical images including brain tumors

2.2.4.1 Features

- Easy to use interface for pictures upload.
- Supports a variety of medical imaging techniques.
- Quick segmentation outcomes.

2.2.4.2 Limitations

- Primarily segmented rather than thoroughly analyzed (progression and recursion)
- Integration of clinical data is limited.
- Professional users do not need verification.

2.2.5 TumorNet

Tumor Net is a deep learning based tumor categorization and detection system.

2.2.5.1 Features

- Uses MRi scans to identify and categorize different type of tumors.
- High glioma detection accuracy.
- Focused on research and containing open-source elements.

2.2.5.2 Limitations

- No analysis of survival or progression.
- Need expensive technology in order to be trained.
- Doesn't have user authentication for moral use

2.3 Comparison between proposed system and existing systems

Table 2.1Comparative Analysis between Proposed system and Existing systems

Sr.	Features	BraTS Based Systems	CBICA Portal	Deep Medic	MedSeg	Tumor Net	LongiTumorSense Proposed System
1.	Tumor Detection	Yes	Yes	Yes	Yes	Yes	Yes
2.	Progression/ Recurrence	No	No	No	No	No	Yes
3.	Survival prediction	No	No	No	No	No	Yes
4.	Ethical Access control	No	No	No	No	No	Yes

2.4 Summary

In this chapter we have focused on existing systems related to brain Tumor detection, we have a brief introduction of each existing system and comparison of existing systems with our developed system. This chapter also highlights the strengths and limitations of existing systems, providing a clear context for our approach. By comparing these studies with our developed system, we aim to demonstrate the improvements our system LongiTumorSense brings to brain tumor detection.

CHAPTER 3
REQUIREMENT SPECIFICATION

3.1 Introduction

Requirements are the main aspect of the system on which the system is built, so requirements should be correct and clear to achieve good results in terms of quality and user satisfaction. Requirements include interface elements that deal with hardware aspects as well as software aspects of the system's requirements, database requirements, functional requirements and non-functional requirements.

This chapter will cover detailed aspects related to requirements stated above as well as use case modeling of the system and use case description for the system. At the end of this chapter, detailed feasibility points are discussed so that project development should be on track.

3.2 Interface Requirements

Following are the interface requirements of our system

3.2.1 User Interface

The usability and design-testing website will have a user-friendly interface to facilitate the users for easy navigation and interaction.

3.2.2 Hardware Interfaces

Computing Device: Users will need access to a computing device such as a desktop, laptop mobile or tablet to interact with the application.

3.2.3 Software Interfaces

The website will have following software interfaces:

- **Web Browser:** The app can be accessed through standard web browsers (Google Chrome, Mozilla Firefox, Safari etc.).
- **Operating System:** The app can be used on common operating systems, including Windows, MacOS and Linux.
- **Development Environment:** Commonly used tools include Visual Studio Code, python, JavaScript.
- **Database services:** The app will utilize a database service for storing user data and assessment results. Commonly used database services may include MongoDB.

3.3 Functional Requirements

Following are the Functional Requirements of LongiTumorSense:

FR-1: LongiTumorSense shall allow new users to create an account.

FR-2: LongiTumorSense shall provide a mechanism for users to reset their password.

FR-3: LongiTumorSense shall allow authenticates users to create multiple patients profiles.

FR-4: LongiTumorSense shall allow users to add clinical data for each patient.

FR-5: LongiTumorSense shall allow users to upload MRI scans for each patient profile.

FR-6: LongiTumorSense shall generate progression analysis report based on MRI scans uploaded for different patients.

FR-7: LongiTumorSense shall allow users to compare MRI scans of different timestamps for the same patient.

FR-8: LongiTumorSense shall be able to save generated reports and allow users to view the history of previously generated reports.

FR-9: LongiTumorSense shall allow users to delete patient's record.

3.3.1 Registration of New User

Table 3.1 illustrates the functional requirement for new User Registration, detailing the process of account creation, email verification and secure user data storage.

Table 3.1 User Registeration

Identifier	FR-1
Title	Registration of new user
Description	User shall be able to create an account as a doctor by providing their personal information and valid medical certifications.
Priority	High

3.3.2 Forget password

Table 3.2 illustrates the functional requirement for forget password, enabling users to reset their password securely.

Table 3.2 Forget password

Identifier	FR-2
Title	Forget password
Description	User shall be able to reset their password by clicking the forget password option, entering their registered email address, accessing the password reset link sent on that email and setting a new password.
Priority	High

3.3.3 Create Multiple Patients Profiles

Table 3.3 illustrates the functional requirement for creation of multiple patients by user (doctor).

Table 3.3 Patient profiles creation

Identifier	FR-3
Title	Create Multiple Patients
Description	User shall be able to create multiple patients profiles to keep the MRI scans of patients organized and maintained for future access.
Priority	High

3.3.4 Add Clinical Data for Patient

Table 3.4 illustrates the functional requirement for adding clinical data of patient by user (doctor).

Table 3.4 Adding Clinical data of Patient

Identifier	FR-4
Title	Adding clinical data for patients
Description	User shall be able to add clinical data for patients.
Priority	High

3.3.5 Upload MRI scans for detection

Table 3.5 illustrates the functional requirement for uploading MRI scans of patient by user (doctor).

Table 3.5 Uploading MRI scans

Identifier	FR-5
Title	Upload MRI's of Patients
Description	User shall be able to upload MRI's of patients for progression/recurrence analysis report.
Priority	High

3.3.6 Generate Progression Report

Table 3.6 illustrates the functional requirement for generating progression analysis report of patient by user (doctor).

Table 3.6 Generate Progression Report

Identifier	FR-6
Title	Generate progression report
Description	User shall be able to generate progression report for patients.
Priority	High

3.3.7 Compare MRI's

Table 3.7 illustrates the functional requirement for comparison of MRI uploads of different timestamps for same patient by the user (doctor).

Table 3.7 MRI's Comparison

Identifier	FR-7
Title	Compare MRI uploads
Description	User shall be able to compare MRI uploads of different timestamps for patients.
Priority	High

3.3.8 Save generated report

Table 3.8 illustrates the functional requirement for saving the generated report of patient as a record to view later.

Table 3.8 Save Generated Report

Identifier	FR-8
Title	Save generated reports and view previously generated reports
Description	User shall be able to save reports generated from MRI's as patient history to view later on.
Priority	High

3.3.9 Delete Patient records

Table 3.9 illustrates the functional requirement for deleting records, allowing users to manage and maintain patient information by removing outdated, redundant or incorrect records, ensuring data accuracy and system cleanliness.

Table 3.9 Logout System

Identifier	FR-9
Title	Delete Patient records
Description	User shall be able to delete patient profiles and their related records to effectively manage outdated or incorrect records.
Priority	High

3.4 Non-Functional Requirements

Following are the non-functional requirements of LongiTumorSense:

3.4.1 Performance

NFR-P1: LongiTumorSense shall respond to user interactions within 3 seconds for 90% of requests under normal operating conditions.

3.4.2 Security

NFR-S1: LongiTumorSense shall authenticate users using encrypted credentials and enforce secure session management to protect user accounts and sensitive data.

NFR-S2: LongiTumorSense shall establish a secure, encrypted communication channel between the application and the database using industry-standard protocols (e.g., TLS) to ensure data confidentiality and integrity.

3.4.3 Availability

NFR-A1: LongiTumorSense shall maintain an operational availability of at least 99% uptime over a 24-hour period, excluding scheduled maintenance.

NFR-A2: LongiTumorSense shall ensure continuous access to core features including brain tumor detection, MRI comparison across different timestamps, and report generation during normal operating conditions.

3.4.4 Usability

NFR-U1: LongiTumorSense shall provide an intuitive user interface that enables first-time users to complete core tasks such as patient creation, MRI upload, tumor detection, and report generation without external training.

NFR-U2: LongiTumorSense shall ensure consistent navigation, clear labeling, and instructional feedback to reduce user errors during system interaction.

3.4.5 Reusability

NFR-R1: LongiTumorSense shall be developed using a modular architecture in which core components (such as authentication, patient management, MRI processing, and report generation) can be reused or extended without significant changes to existing code.

NFR-R2: LongiTumorSense shall allow integration of additional analysis modules or AI models with minimal modification to the existing system structure.

3.4.6 Flexibility

NFR-F1: LongiTumorSense shall support modification and addition of system features through its modular architecture without requiring major changes to existing components.

NFR-F2: LongiTumorSense shall allow integration of new analytical or AI modules without affecting the core patient management and data storage functionalities.

3.4.7 Maintainability

NFR-M1: LongiTumorSense shall allow developers to fix reported bugs, apply updates, or add new features within a reasonable time without affecting existing functionality.

NFR-M2: LongiTumorSense shall have a well-structured and modular codebase with clear documentation covering all modules to support easy troubleshooting and long-term system stability.

3.4.8 Portability

NFR-P1: The system shall be compatible with major web browsers, including Chrome, Firefox, Edge, and Safari**,** without loss of functionality.

NFR-P2: The system shall provide a responsive user interface that functions correctly on desktop, tablet, and mobile devices**,** maintaining consistent layout and usability.

3.5 Use Case Modeling

A complete use case modeling is presented below in Figure 3.1 with the help of use case diagram.

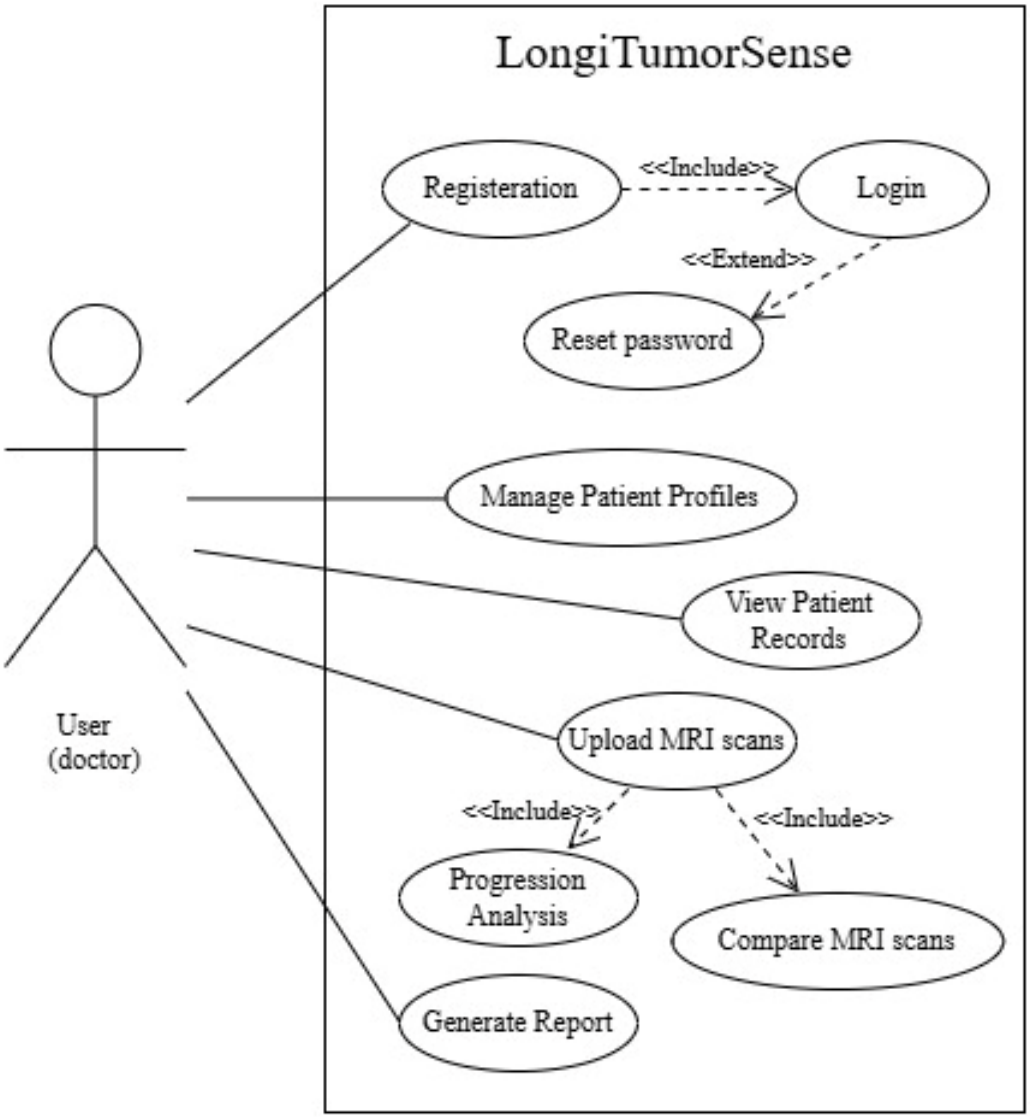


Figure 3.1 System Use Case Diagram

3.6 Use Case Description

Following are the fully elaborated use cases of the system:

3.6.1 Use case: User Registration

Table 3.10 provide detailed information regarding the registration process.

Table 3.10 Use case description for User Registration

Use case id	UC-01
Name	User Registration
Actors	Doctor
Pre-condition	The user is not already registered and user must belong to medical profession to register
Post-condition	The user account has been created successfully
Main Success Scenario	
Actor Action	**System Response**
1. The user clicks the Register Button	1.1. The system will proceed with the user request and open the sign-up page.
2. The user fills the necessary information and submits it.	2.1. The system will validate the provided information and user medical documents would be validated to ensure that user belongs to medical profession. After validation, the user account would be created successfully.
Alternative Path	
Actor Action	**System Response**
1. User enters invalid email format.	1.1. If the email format is invalid, the system will prompt the user to enter a valid email address.
2. The user missed out on any mandatory text field on a form and clicked the submit button	2.1. If the user missed any mandatory text, field system will prompt the user to enter the required information.
3. User entered a password that does not meet the password security requirements.	3.1. If a user entered password other than the valid password format like all lower case alphabets, underscores and digits. System Prompts the user to enter a valid format

3.6.2 Use case: Log In

Table 3.11 highlights the use case description of the Login Process, providing detailed information regarding the user's action and system response. It also outlines the sequence of steps involved, ensuring clarity in understanding the login workflow.

Table 3.11 Use case description for User Login

Use case id	UC-02
Name	Log In
Actors	Doctor
Pre-condition	The user must be registered and have valid credentials.
Post-condition	The user successfully logged into the system.
Main Success Scenario	
Actor Action	**System Response**
1. User enters valid credentials	1.1. The system verifies the credentials per criteria.
2. The user clicks the login button	2.1. The system will provide access to the user once they are logged in.
Alternative Path	
Actor Action	**System Response**
1. The user entered an incorrect username or password.	1.1. If the email format is invalid, the system will prompt the user to enter a valid email address.
2. If the User forgets the password, they can select forgot password option	2.1. If the user missed any mandatory text, field system will prompt the user to enter the required information.

3.6.3 Use case: Reset password

Table 3.12 shows the use case description for resetting the account password, detailing the user's interaction with the system during the password reset process. It also outlines the steps involved in verifying the user's identity for security purposes. Additionally, the table explain the system's response after successful password reset, ensuring a seamless experience for the user.

Table 3.12 Use case description for Password reset

Use case id	UC-03
Name	Reset Password
Actors	Doctor
Pre-condition	The user must have a registered account and provide the registered email with his/her account.
Post-condition	Password reset successfully.
Main Success Scenario	
Actor Action	**System Response**
1. The user clicked the forget password option and provided the associated email with his/her account to reset account password.	1.1. The system verifies the email or registered number by sending a verification code or link to reset password.
2. User enter the code sent to their email by system and proceed.	2.1. Once the code is verified, system allows the user to create new password.
3. User created a new password and saved to proceed.	3.1. System updates the password to the new password and notifies that password has been changed successfully.
Alternative Path	
Actor Action	**System Response**
1. User enters invalid email or number.	1.1. System will prompt the user to enter valid email or number.

3.6.4 Use case: Manage Patients Profile

Table 3.13 shows the use case description for creation of patients by the doctor so that they can upload MRIs and manage each patient data separately. Patient's clinical information such as age, gender and other relevant information would be required to create patient.

Table 3.13 Use case description for Patient creation

Use case id	UC-04
Name	Manage Patient Profiles
Actors	Doctor
Pre-condition	The user(doctor) must be login through his/her registered account to create and manage patients profile.
Post-condition	Patient profile created successfully.
Main Success Scenario	
Actor Action	**System Response**
1. The user(doctor) clicked the create patient button to create a new patient profile.	1.1. The system asks for clinical data of the patient to be created.
2. User enters the clinical data related to the patient to create patient profile.	2.1. System verifies the provided clinical data of the patient, if it fulfills the required clinical data criteria the patient profile would be created.
3. User (doctor) saves the patient data to finish profile creation.	3.1. System saves the clinical data of the patient and creates a new patient profile that would be visible to doctor that created profile.
Alternative Path	
Actor Action	**System Response**
1. User (doctor) enters invalid clinical data.	1.1. System will prompt the user to enter clinical data of the patient as per requirements.

3.6.5 Use case: View Patient Medical Records

Table 3.14 shows the use case description for viewing the patient medical records, detailing the steps involved in accessing and reviewing a patient medical history. It outlines the doctor actions to search for and select the relevant patient records, as well as the systems response in retrieving and displaying the data. The table also highlights security measures to ensure that only authorized user can access the records.

Table 3.14 Use case description for Medical Records History

Use case id	UC-05
Name	View Patient Medical Records
Actors	Doctor
Pre-condition	The user(doctor) must be logged into the system.
Post-condition	The user(doctor) successfully views the patient previous records of MRI scans and their reports.
Main Success Scenario	
Actor Action	**System Response**
1. The user(doctor) selects patient to view his/her Patient History.	1.1. The system will proceed with the user request and display the required interface upon user request.
2. The doctor selects the add or remove patient from the patient list.	2.1. System will proceed with the users request and add or remove the patient from the patient list.
3. If a report is available Doctor selects the view report option.	3.1. System will display the patient report if it is completed and available to display.
Alternative Path	
Actor Action	**System Response**
1. Doctor clicks to view a report that is in progress	1.1. If the report is in progress system will ask the Doctor to wait until the report is created completely and will be available to view.

3.6.6 Use case: Upload MRI scans

Table 3.15 shows the use case description for uploading the MRI scans of the patient for the purpose of Brain Tumor Detection.it Outlines the user (doctor) actions such as, selecting MRI's to upload for detection purposes and system response to initiate the scanning process. The table also highlights the image processing steps involved and how the system analyzes the images to detect Brain Tumors. This use case ensures efficient interaction and accurate detection during the MRI scans analysis.

Table 3.15 Use case description for MRI Upload

<table>
<tr><td>Use case id</td><td colspan="2">UC-06</td></tr>
<tr><td>Name</td><td colspan="2">Upload MRI scans</td></tr>
<tr><td>Actors</td><td colspan="2">Doctor</td></tr>
<tr><td>Pre-condition</td><td colspan="2">The user(doctor) must have MRI scans in local storage of his device to upload and detect brain tumor.</td></tr>
<tr><td>Post-condition</td><td colspan="2">MRI scans have been scanned successfully and report is generated.</td></tr>
<tr><td colspan="3">Main Success Scenario</td></tr>
<tr><td colspan="2">Actor Action</td><td>System Response</td></tr>
<tr><td colspan="2">1. The user(doctor) clicked the button to upload MRI scans from the local storage.</td><td>1.1. The system saves the uploaded MRI scans and wait for user action to either compare or detect tumor.</td></tr>
<tr><td colspan="2">2. The doctor selects the option for either Comparison when 2 MRI uploads are added or either for tumor detection when a single MRI is uploaded.</td><td>2.1. System will proceed for generation of tumor detection report when single MRI is uploaded and when multiple MRIs are uploaded system will compare to generate progression report</td></tr>
<tr><td colspan="3">Alternative Path</td></tr>
<tr><td colspan="2">Actor Action</td><td>System Response</td></tr>
<tr><td colspan="2">1. Doctor clicks to generate report or comparison without uploading MRI scans.</td><td>1.1. System will response to upload MRI scans before clicking Tumor detection or comparison button.</td></tr>
</table>

3.7 Resource Requirement

Following are the resources required for the system to work efficiently:

3.7.1 Hardware Requirements

Following hardware requirements provide an overview of the essential resource required for the development of the system. Table 3.16 lists Hardware Requirements.

Table 3.16 System Requirements

Resources	Requirements	Rationale
Laptop/PC	Corei5 5th gen , 8gb Ram or above	For the development and maintenance of the system
Internet	4g or above	For the development of the system as well as basic guidelines and help
Android Mobile/iOS	Android version 5.0 or iOS9	For emulation

3.7.2 Tools/Technology

Following software tools and technologies will be required for the development of LongiTumorSense Brain Tumor detection system.

- Python (PyTorch)
- React JS
- Express Js
- BraTS-Style U-Net,3D U-Net, Enhanced 3D U-Net, DenseNet-121 , ResNet-50, EfficientNet-B4
- Jupyter Notebook/Google Colab
- NIfTI(.nii),CSV, PNG, JPEG
- Pandas,NumPy
- Matplotlib, Seaborn,ITK-SNAP
- NIBabel,SimpleITK

3.8 Human Effort

Table 3.17 shows the human effort for the system.

Table 3.17 Human Effort

Task	Abdullah	Hassan Ali	Ehtasham Ul Haq
Problem Identification	✓	✓	✓
Report Writing	✓	✓	✓
Requirements Gathering	✓	✓	✓
Interface Designing	✓	✓	✓
Coding	✓	✓	✓
Database Connectivity	✓	✓	✓
Testing	✓	✓	✓

3.9 Database Requirements

The system will use MongoDB as its database, making sure that user data is kept private and secure. Mongoose flexible structure and powerful search features will help manage data effectively. Users and service providers will need a steady internet connection to access the system and its database.

3.10 Summary

The chapter covers the details regarding the requirement specification and user interactions with the system, through use case diagram and detailed use cases. A slight overview of the resource requirement is given providing details about what type of hardware will be required and what kind of tools and technology will be required to develop the system.

CHAPTER 4
SYSTEM MODELING

4.1 Introduction

This chapter focuses on the modeling of the system under development. System modeling involves creating an abstract representation of the system, with each model offering a unique perspective on the system. Utilizing the Unified Modeling Language (UML) notation, we have provided a graphical depiction of the system for a better understanding of the system's functionalities and communication methods with the user.

4.2 (4+1) View Model

For better communication, planning and Implementation of our system, we have adopted (4+1) view model for our system to describe its architecture in a comprehensive and organized manner. This model helped us to capture different aspects of the system through each view, Figure 4.1 shows the (4+1) View Model:

1. **Logical view:** illustrated using UML Class Diagram
2. **Process view:** Represented using UML Sequence and Activity Diagram.
3. **Development view:** Represented using UML Component Diagram.
4. **Physical view:** illustrated using the Deployment Diagram.

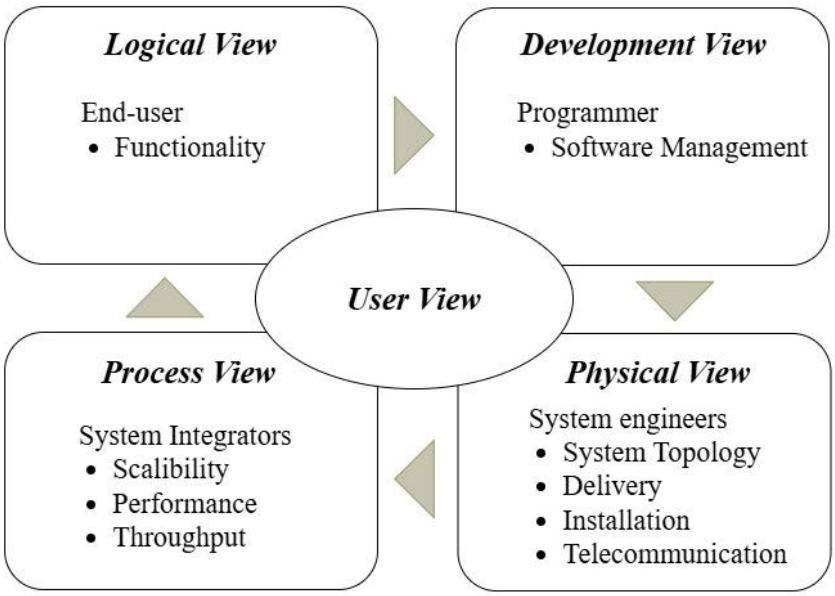


Figure 4.1 (4+1) View Model

4.3 Logical View

In logical View, high-level organization of our system is demonstrated, we have shown relationships and hierarchies between different components and entities within the architecture utilizing the class diagram.

4.3.1 Class Diagram

The Class Diagram describes the structure of the system by showcasing classes, attributes, methods and relationship among objects, Figure 4.2 illustrates the class diagram of our system highlighting the specific classes involved, their characteristics and the nature of their connections to one another.

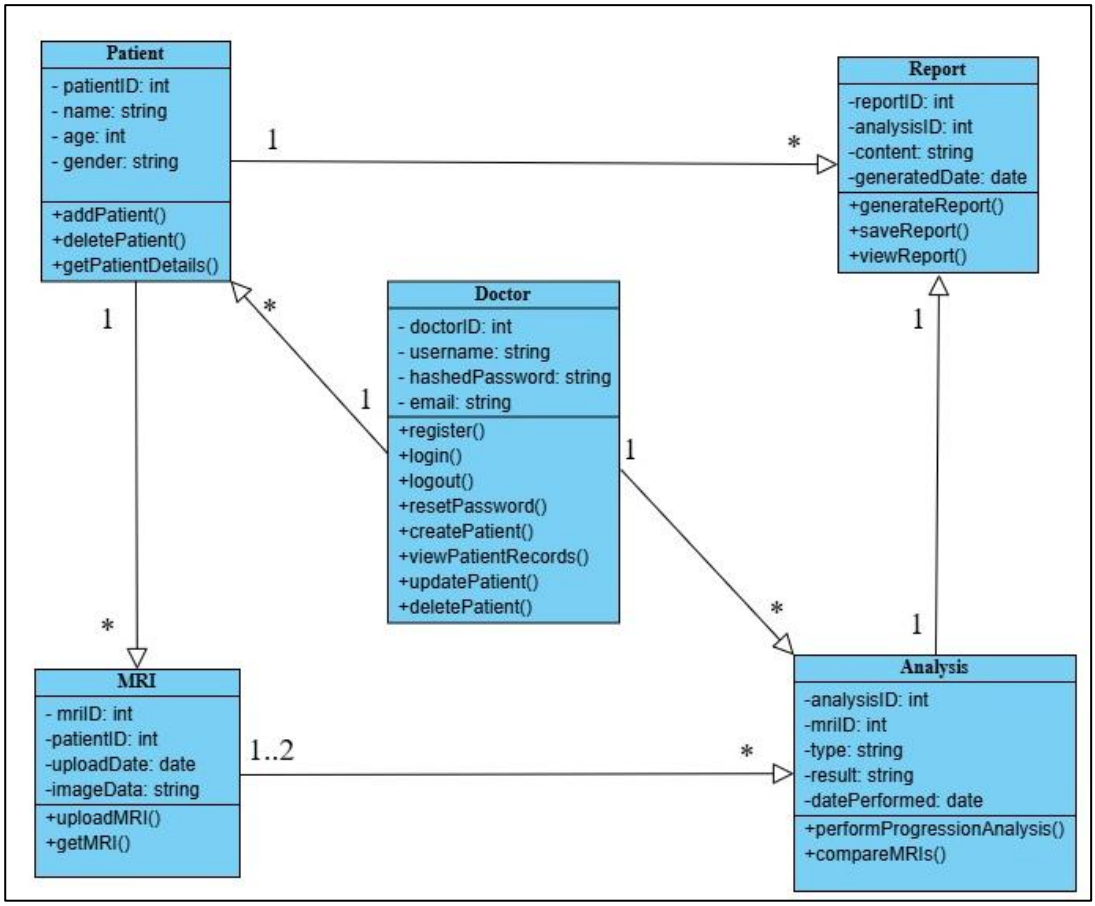


Figure 4.2 System Class Diagram

4.4 Process View

Process View focuses on the dynamic aspect of the system, highlighting how different component of the system interact and collaborate to achieve the desired functionality. Folr the process view we have used sequence diagram and activity diagrams to illustrate the flow of our system.

4.4.1 System Activity Diagram

Figure 4.3 shows the workflow and activities of the doctor within the system, visually representing their interactions and activities. It also highlights the key processes such as patient management, reviewing medical records and generating reports. The diagram provides a clear view of doctor's role, ensuring a better understanding of their activities within the system.

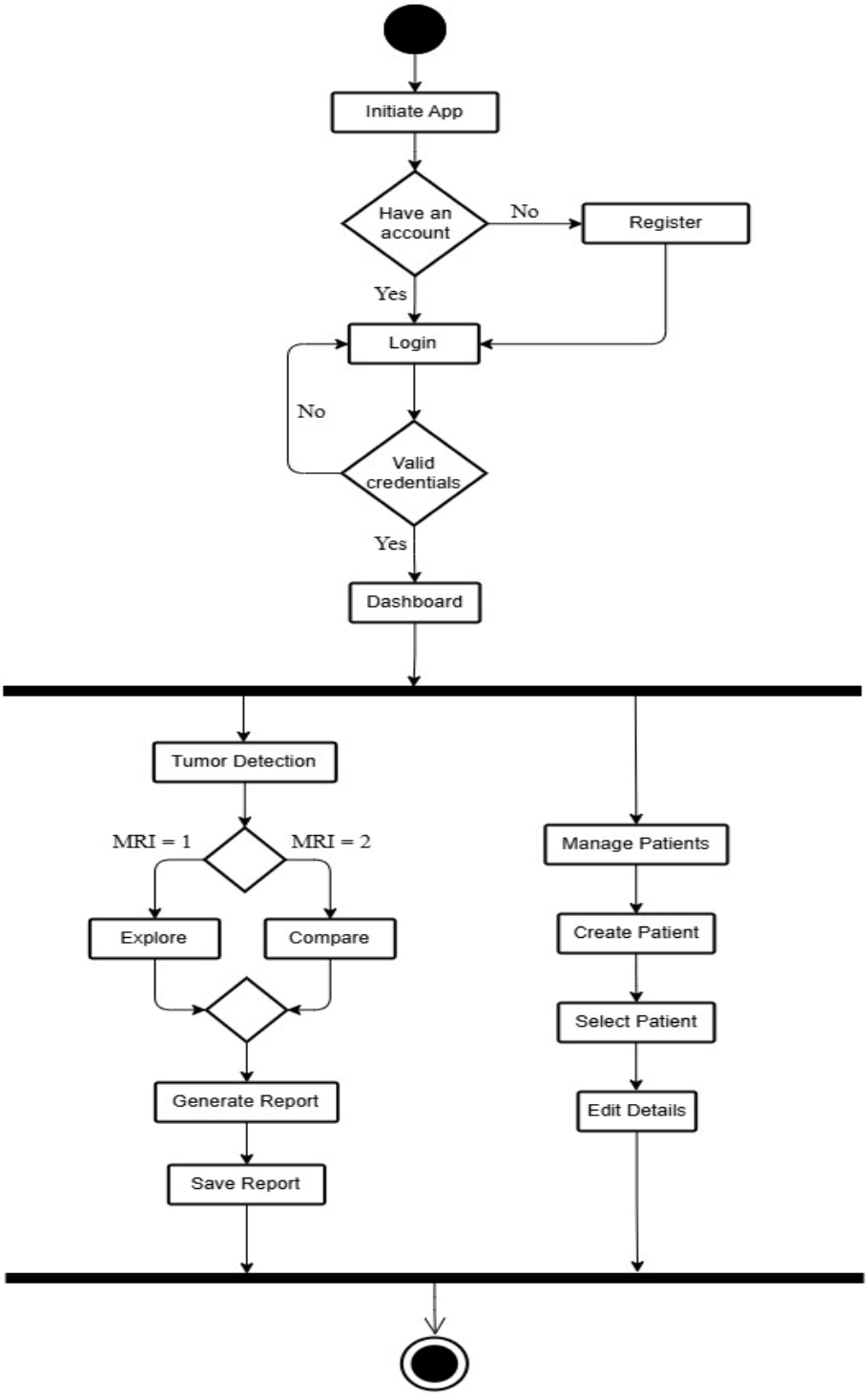


Figure 4.3 System Activity Diagram

4.4.2 Sequence Diagram

Sequence Diagram illustrates the sequence and flow of control of various objects visually, showing how objects collaborate to achieve different functionalities. These diagrams provide a clear view of dynamic behavior of the system and interaction between different components. Following are the sequence diagrams of our system, which highlights various processes and interactions in detail.

4.4.2.1 Registration Sequence Diagram

The given Figure 4.4 represents the user registration process in the system. It illustrates the interaction between the user, the registration module and admin approval for account creation detailing the step-by-step flow of actions. The process is initiated when doctor clicks on “Sign up” button leading towards the form. The system validates the details either they are valid or need correction. If incorrect details are entered doctor is prompted to re-fill the wrong fields and enter correct details. Successful registration ensures the creation of account and storage of information in database for future validation of credentials.

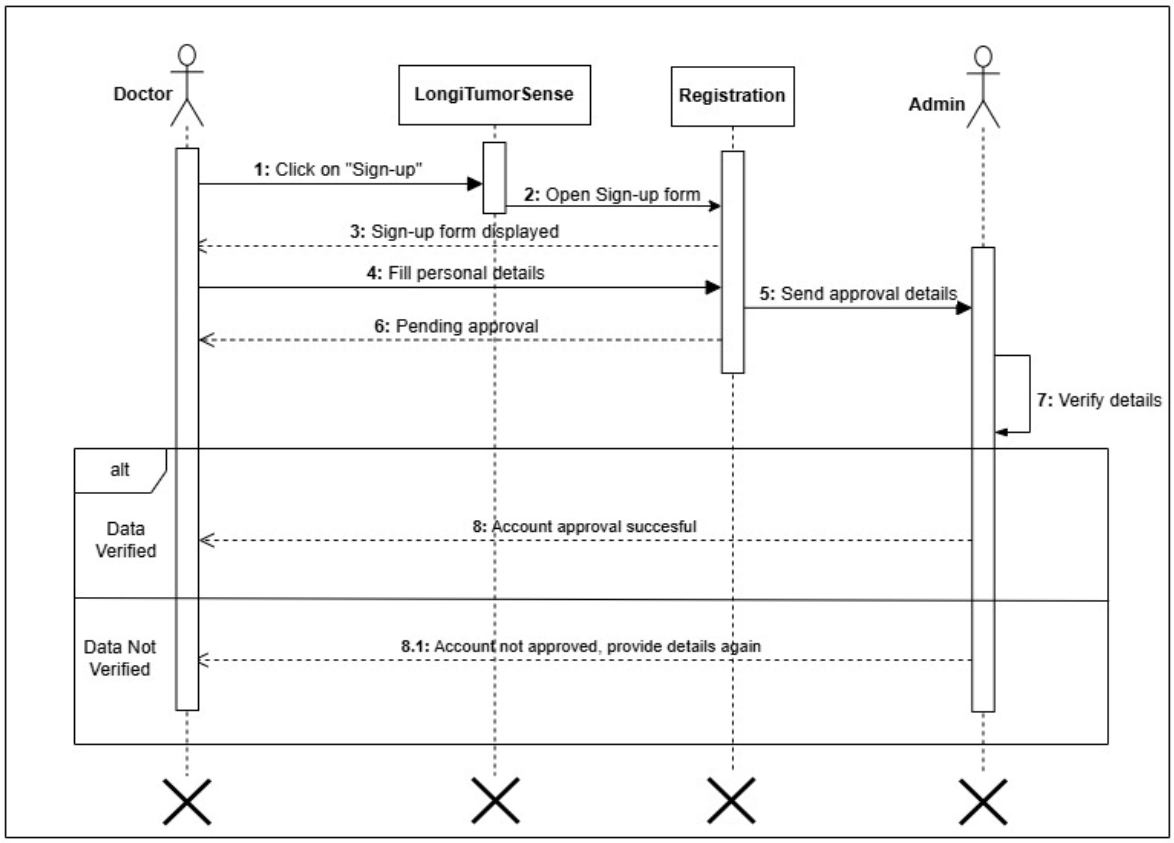


Figure 4.4 Registration Sequence Diagram

4.4.2.2 Login Sequence Diagram

The given Figure 4.5 represents the process when user login the system, visually representing the sequential steps and interactions involved in the process. It illustrates interaction between the user, login form and the database, highlighting how these components work together to validate user credentials and grant access to the system. The process starts when user enter his credentials. The system then validates the details against the credentials stored in the database, checking for correction. If details are valid user's login successful, otherwise they are prompted to enter credentials again that are valid. This structured approach ensure secure authentication and prevents unauthorized access to the system.

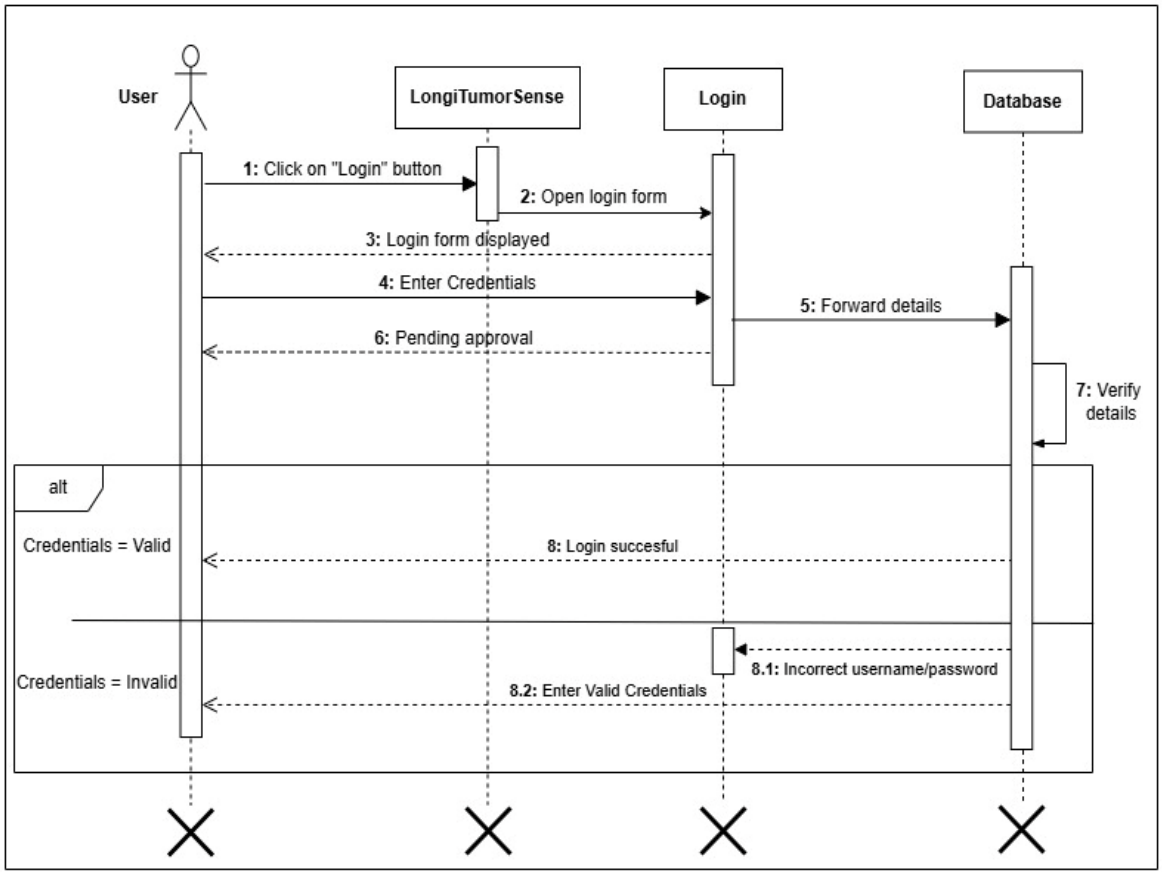


Figure 4.5 Login Sequence Diagram

4.4.2.3 Explore Sequence Diagram

The given Figure 4.6 represents the process when doctor uploads MRI to generate report through the explore module of system. When doctor want to generate the Analysis report firstly, he will create a new patient with his clinical details added or will select an existing patient to use clinical data of the patient. After selecting patient, doctor will upload MRI from device and MRI file would be validated for correct format if it is valid the Analysis report would be generated otherwise system will prompt doctor to upload a valid format MRI.

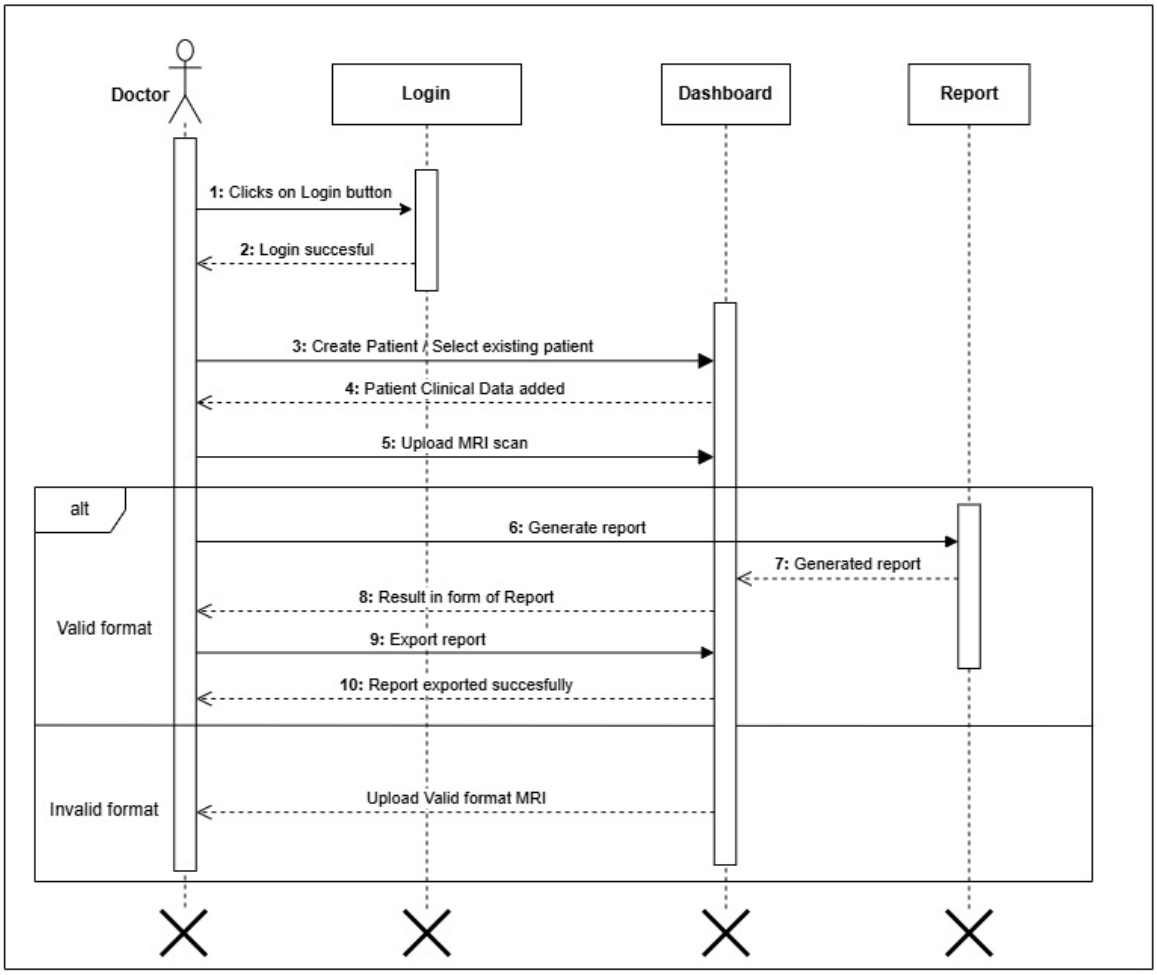


Figure 4.6 Explore Sequence Diagram

4.4.2.4 Compare Sequence Diagram

The given Figure 4.7 represents the process when doctor uploads MRI to generate comparison report through the system. When doctor want to generate the Comparison report firstly, he will create a new patient with his clinical details added or will select an existing patient to use clinical data of the patient. After selecting patient, doctor must upload 2 MRI's from device for comparison. The comparison report would be generated only when doctor uploads 2 MRI's and their format should be valid.

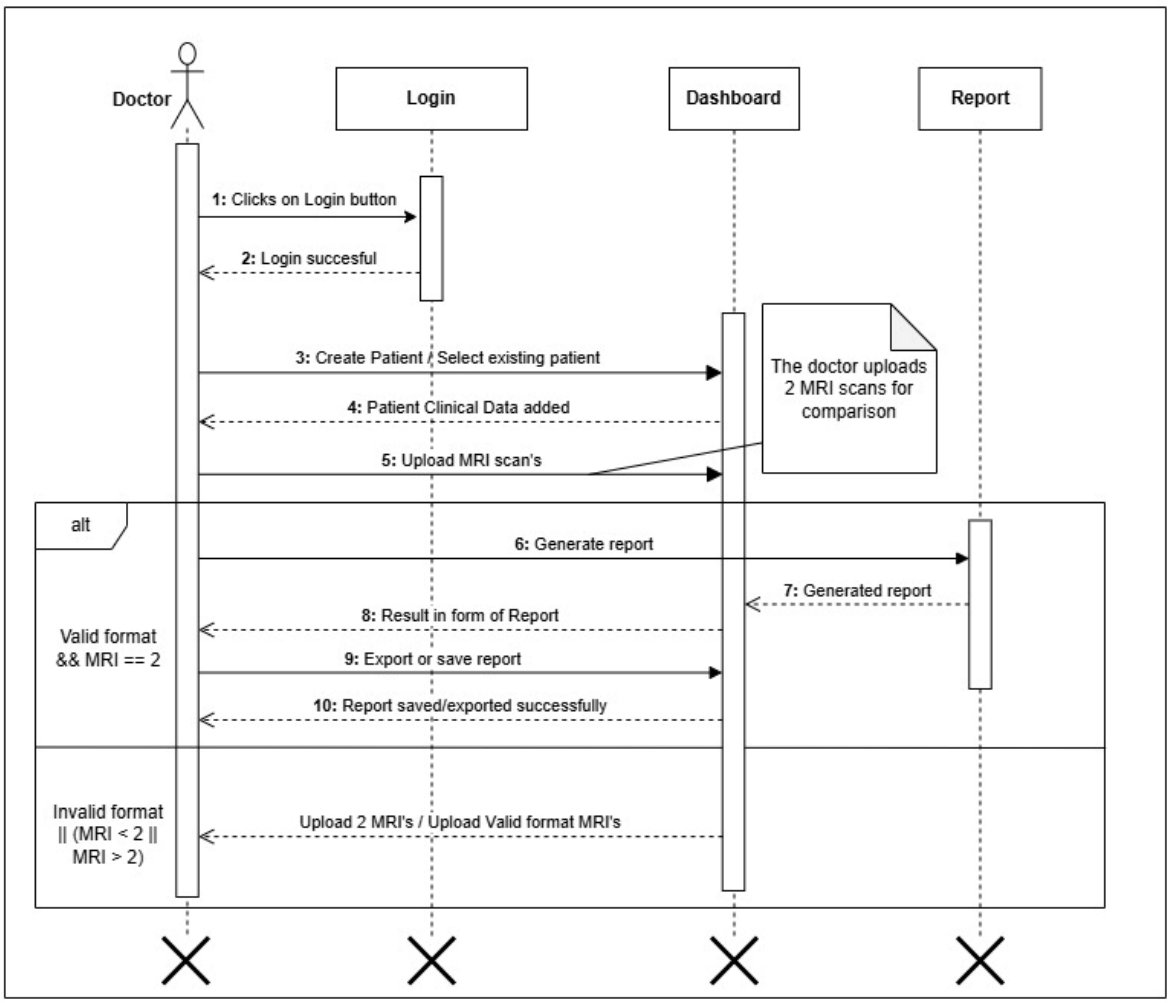


Figure 4.7 Compare Sequence Diagram

4.4.3 ER Diagram

The given Figure 4.8 represents the ER diagram of LongiTumorSense, demonstrating how different entities interact to support secure access, patient management and AI-based tumor analysis. The ER diagram helps in modeling the core data elements involved in the tumor detection process, including authorized medical professionals, MRI scans and generated analysis reports. Although the system is implemented using a NoSQL database, the ER diagram serves as a conceptual representation of the data model, highlighting logical relationships rather than physical storage constraints.

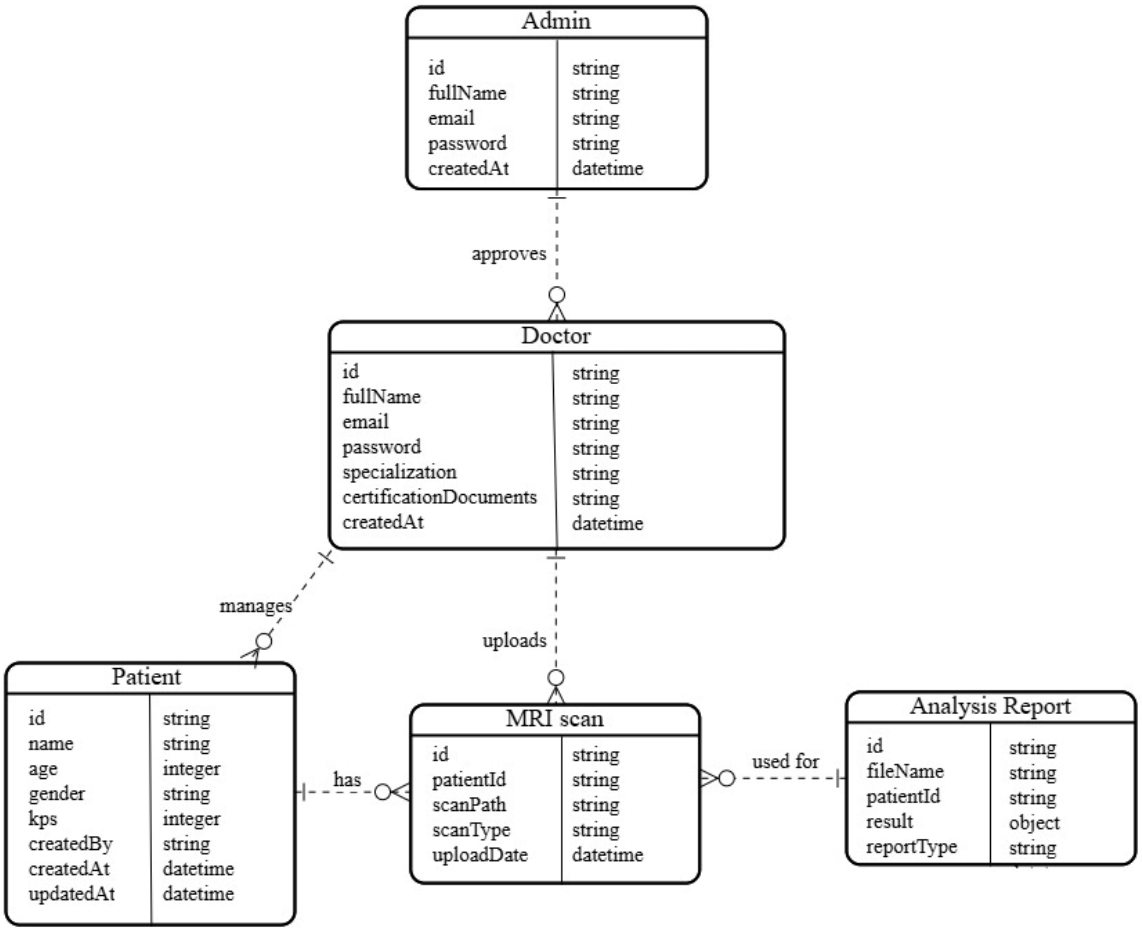


Figure 4.8 Entity Relationship Diagram for LongiTumorSense

4.5 Development View

The development view of our system depicts using the UML component diagram, which focuses on the structure and organization of the software uder development. This diagram highlights the major components, their relationships and how they interact with each other to fullfill system requirements. It provides a clear representation of the system's modular architecture. Additionally, it emphasizes the dependencies between components, facilitating better understanding and management of the system's development process.

4.5.1 Component Diagram

Figure 4.9 highlights the visual representation of the system components and their relationship with each other, providing a blueprint of the system's structure.

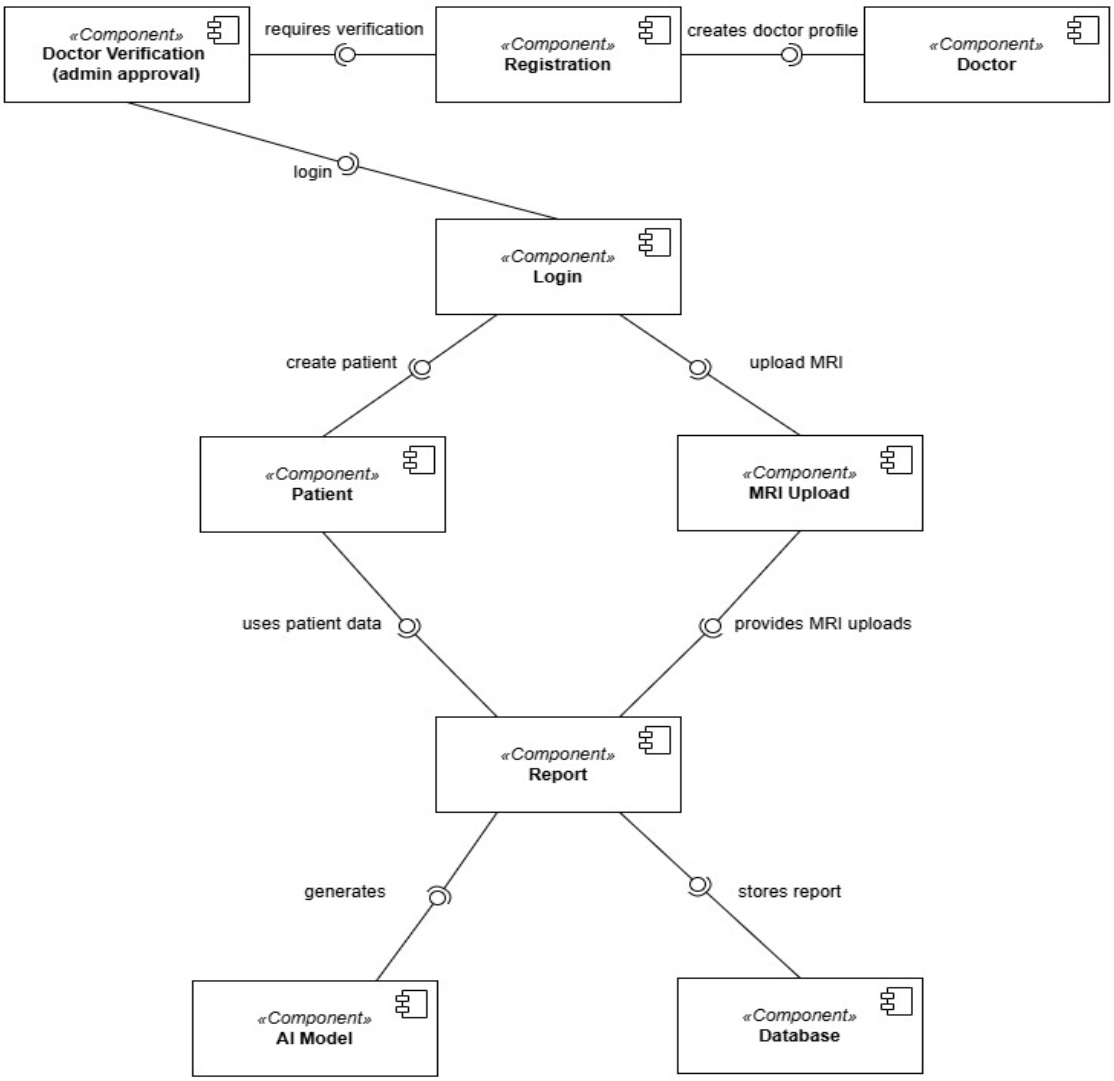


Figure 4.9 System Component Diagram

4.6 Physical View

Physical view shows the deployment perspective of the system, providing the details how our software system is mapped on hardware components and their relationships focusing on where the system executes.

4.6.1 Deployment Diagram

The following deployment diagram depicts our system components and where they are deployed. Figure 4.10 represents our system's deployment diagram.

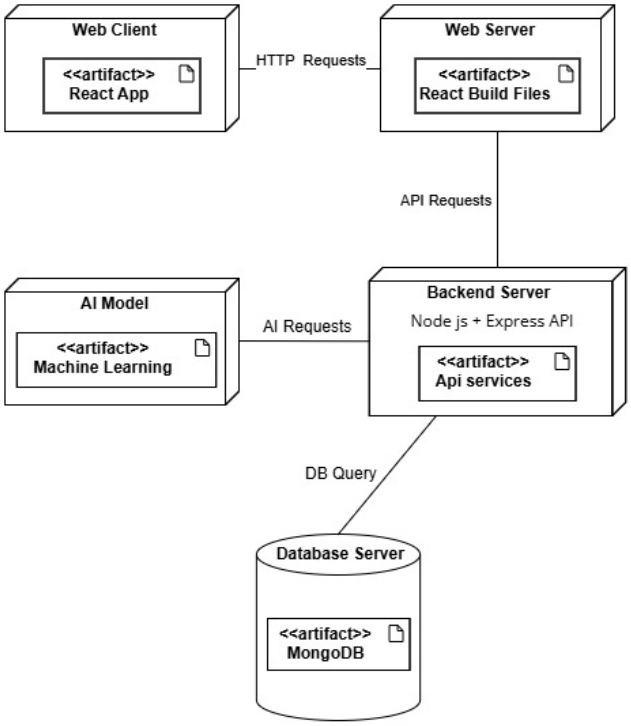


Figure 4.10 Deployment Diagram

4.7 System Prototype

Figure 4.11 shows the Landing page

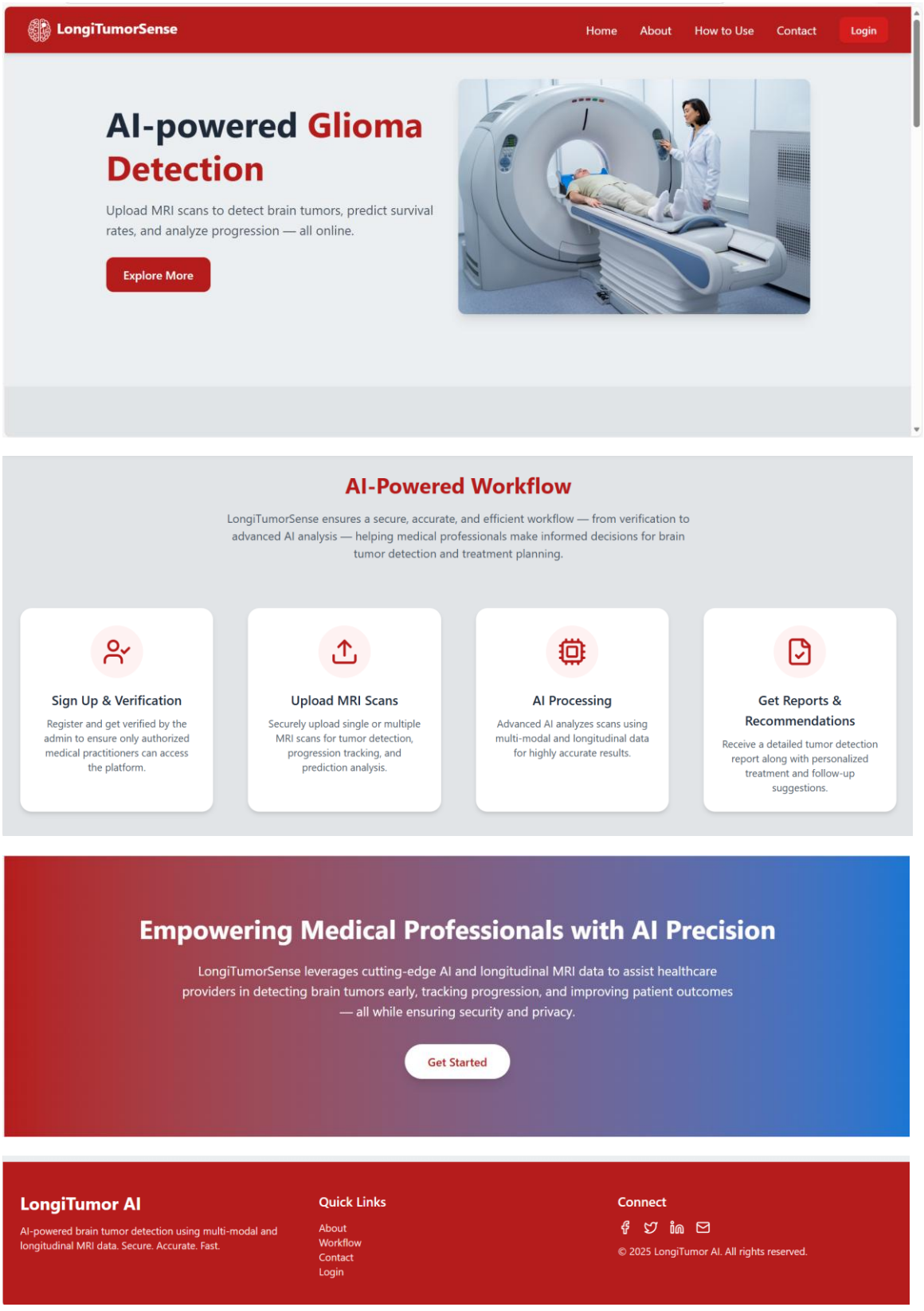


Figure 4.11 System Landing page

Figure 4.12 shows the Login screen.

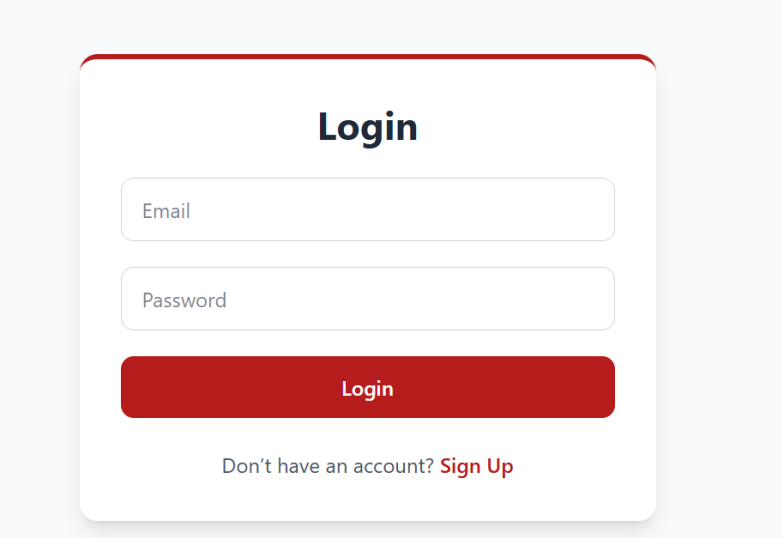


Figure 4.12 Login screen

Figure 4.13 shows the Registration screen.

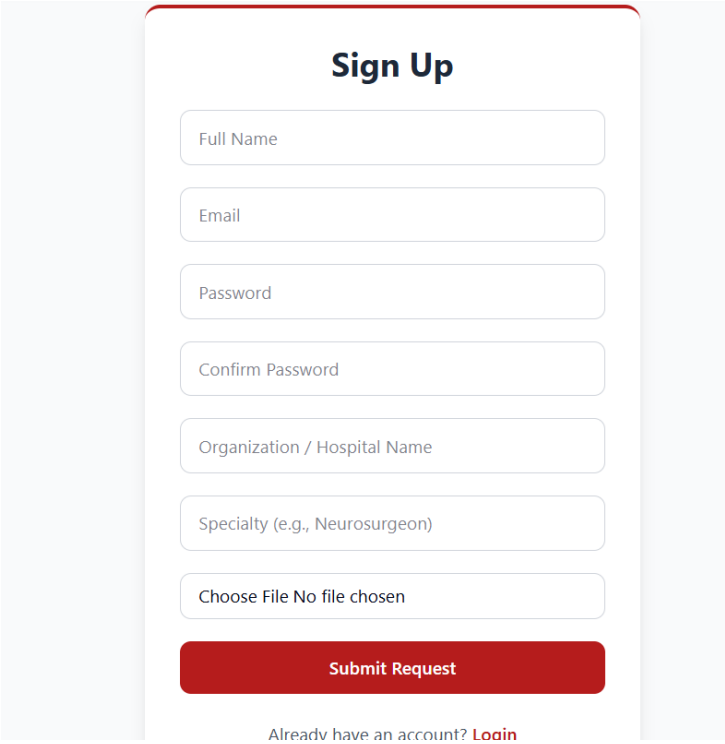


Figure 4.13 Registration screen

Figure 4.14 shows the Doctor Interface.

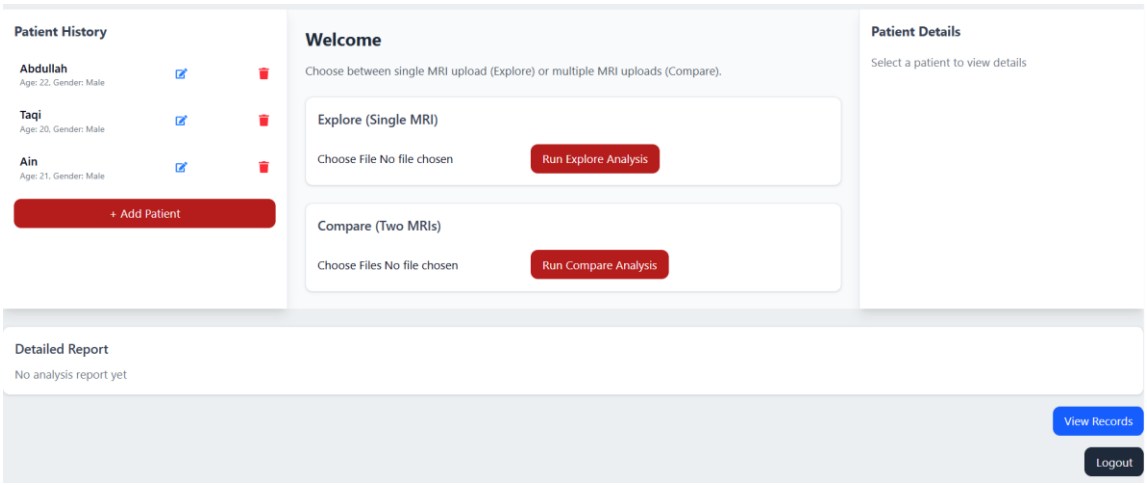


Figure 4.14 Doctor Interface

Figure 4.15 shows the Admin Interface.

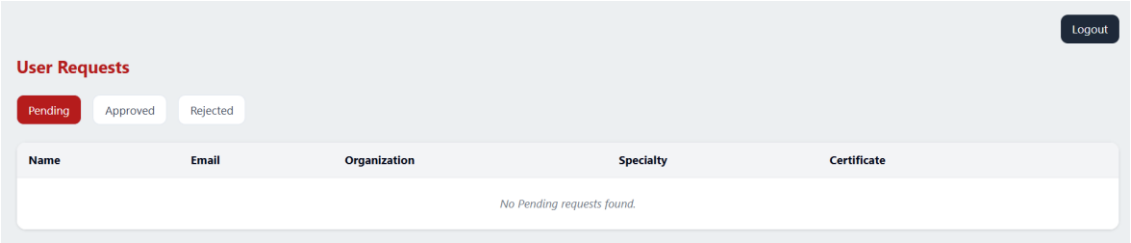


Figure 4.15 Admin Interface

Figure 4.16 shows adding Patient Clinical Data.

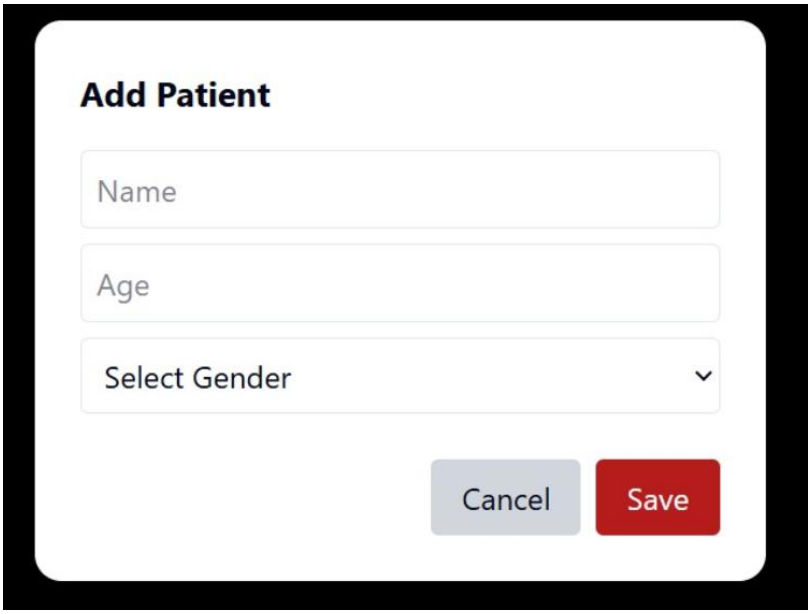


Figure 4.16 Adding Patient Clinical Data

Figure 4.17 shows Generated Report.

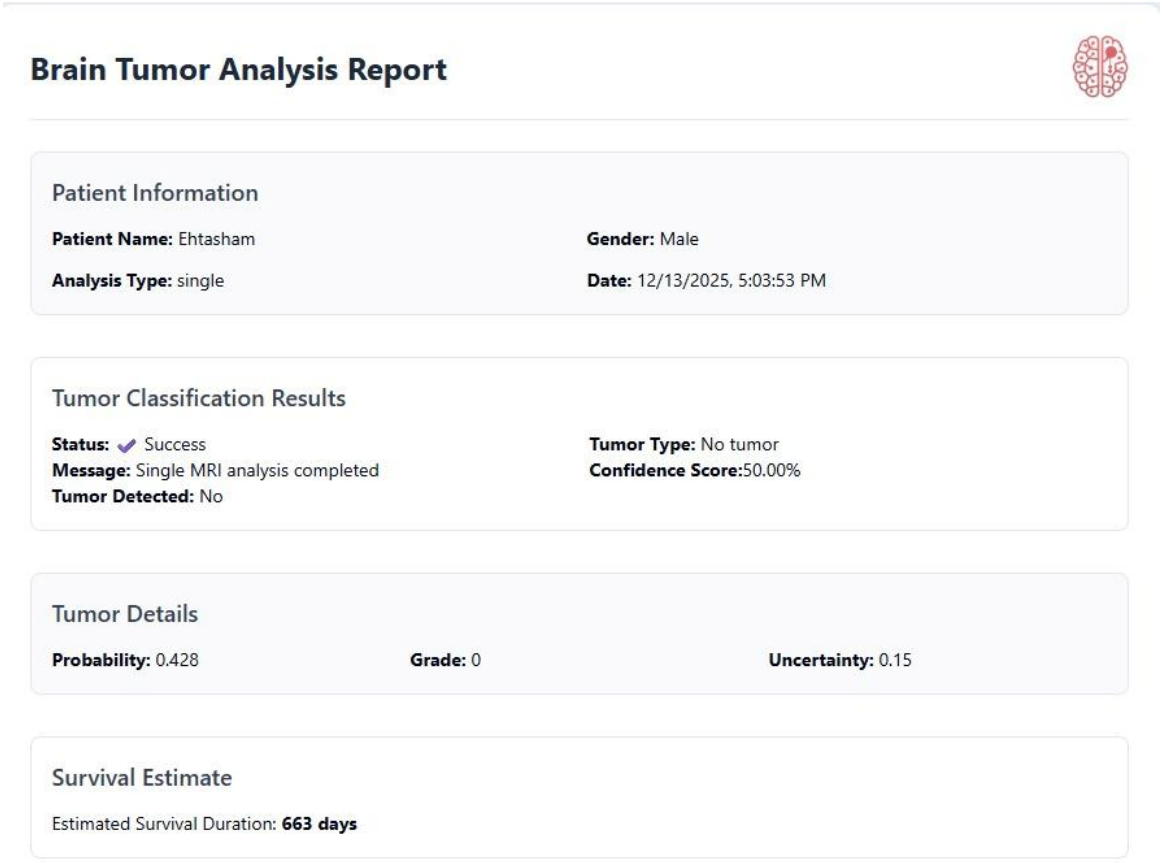


Figure 4.17 Generated Report

Figure 4.18 shows Treatment Recommendations.



Figure 4.18 Treatment Recommendation

4.8 Summary

This chapter provides detailed UML diagrams of our system. It covers the overall system design, design approaches, interface design of the system, 4+1 View model, representing logical view using class diagram, process view using sequence diagram and physical view using deployment diagram.

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